

Hidden Gender Gaps in Child Poverty

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September 2026

ABSTRACT. While son preference in India is well documented, measuring gender differences in children's consumption remains challenging because consumption is typically observed only at the household level. We estimate gender-specific child consumption using a collective household model and data from rural India. We find substantial within-household inequality: boys consume systematically more household resources than girls. Translating these allocations into individual-level poverty, roughly 63.6 percent of girls are poor, compared to 48.4 percent of boys. These gaps account for age- and sex-specific needs, persist across most of childhood, and are largely invisible to standard household-level poverty measures. We further show that son-biased fertility stopping behavior contributes to the gender gap in child poverty by increasing girls' exposure to larger families with fewer resources per child, explaining less than 10 percent of the overall gap. Finally, combining our model-based poverty estimates with administrative mortality data, we show that hidden gender gaps in child poverty can account for one in three Indian missing girls.

JEL codes: D13, I32, J13, J16

Keywords: child poverty, gender gap, fertility, intra-household allocation, missing women

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1 Introduction

Gender inequality in economic opportunity may begin within the family. A family’s resources need not be equally available to all of its children. If girls systematically command fewer resources than boys, intrahousehold inequality may shape both current material deprivation and the health and human-capital investments through which childhood conditions affect later outcomes (Almond et al., 2018). Yet these differences are difficult to detect using standard measures of poverty, which typically observe consumption only at the household level and therefore cannot reveal how resources are distributed among individual members.

This measurement problem is particularly consequential in India, where son preference is well documented (Jayachandran, 2015, 2023). Son preference shapes fertility decisions, survival, health investments, and human-capital accumulation, and contributes to the phenomenon of *missing women* (Sen, 1990; Coale, 1991; Anderson and Ray, 2010, 2012).¹ A large literature documents gender disparities in specific investments and outcomes. Girls have historically received fewer vaccinations than boys (Oster, 2009), are breastfed for shorter durations (Jayachandran and Kuziemko, 2011; Barcellos et al., 2014), receive less childcare time and vitamin supplementation (Barcellos et al., 2014), and face differential survival and sex-selection risk (Bhalotra and Cochrane, 2010; Anukriti et al., 2022). Much less is known, however, about whether these disparities emerge in the everyday allocation of household consumption. In particular, do boys receive a larger share of household consumption than girls, and does unequal allocation translate into higher poverty rates among girls?

This paper answers these questions by estimating gender-specific consumption allocations within households in rural India. We use data from the Rural Economic and Demographic Survey (REDS), which records clothing and footwear expenditures separately for adult men, adult women, boys under 16, and girls under 16. Following Dunbar et al. (2013) and Calvi (2020), we combine these assignable expenditures with a collective household model to recover *resource shares*—the fraction of *total* household expenditure allocated to each household member—in both nuclear and extended families, while explicitly distinguishing between boys and girls. We then use the estimated resource shares to construct individual-level consumption and poverty measures for children by gender.

The basic intuition of the approach is simple. Consider two otherwise similar households that differ in total expenditure. If spending on boys’ clothing rises more steeply with total household expenditure than spending on girls’ clothing, this differential response contains information about the relative resources allocated to boys and girls. Under the preference restrictions developed by Dunbar et al. (2013), the slopes of boys’ and girls’ clothing Engel curves are proportional to their respective resource shares. Because resource shares across household members must sum to one, comparing these slopes across men, women, boys, and girls allows us to recover how total household resources are divided among them. Importantly, identification comes from Engel-curve slopes rather than from levels of clothing expenditure, which may reflect differences in tastes or other determinants of clothing demand.

¹The term *missing women* refers to the shortfall in the number of surviving females relative to the number that would be expected in the absence of gender discrimination, given biologically plausible sex differences in mortality.

Our estimates reveal substantial gender inequality in the allocation of household resources to children. In the full sample, which includes both nuclear and extended families, boys receive on average 15.5 percent of household resources per child, compared with 11.7 percent for girls. The disparity is considerably larger in nuclear families, where the girl–boy gap widens to 7.0 percentage points. Adult allocations are also highly unequal, with men receiving substantially larger per-person shares than women, consistent with previous evidence on adult intrahousehold inequality in India (Calvi, 2020; Calvi and Keskar, 2021; Bandyopadhyay and Maity, 2026; Kapoor and Ravi, 2025).

To assess the welfare implications of unequal allocation, we translate estimated resource shares into child-level consumption and poverty. For each child, model-implied consumption equals the estimated per-person resource share multiplied by total household expenditure. Unlike the conventional per-capita approach, which assigns the same level of consumption and poverty status to all household members, our framework allows boys and girls in the same household to have different consumption levels and poverty outcomes (Dunbar et al., 2013; Brown et al., 2021, 2022).

A child is classified as poor when estimated consumption falls below a child-specific poverty threshold. Our preferred measure rescales the adult poverty benchmark using age- and sex-specific caloric requirements; for comparison, we also use the equivalence-scale adjustment commonly employed in the literature, which assigns children under age 15 a threshold equal to 60 percent of the adult benchmark. Both adjustments account for lower resource needs among children, while the calorie-based measure additionally allows needs to vary by age and sex.

Under both measures, girls experience substantially higher poverty than boys. Using the calorie-adjusted thresholds, roughly 63.6 percent of girls are poor, compared with about 48.4 percent of boys. Under the equivalence-scale adjustment, 53.8 percent of girls and 33.5 percent of boys are poor. These disparities persist across most of childhood and are largely invisible to standard household-level poverty measures.

Unequal sharing also implies that poverty may be hidden within households classified as non-poor. Among children living in households whose per-capita expenditure exceeds the conventional poverty line, nearly 42.3 percent of girls and 28.8 percent of boys are nevertheless individually poor under the calorie-adjusted measure. We refer to these disparities as *hidden gender gaps in child poverty*: gender differences in deprivation that are obscured when poverty is measured only at the household level.

We next ask how much of the gender gap reflects differential treatment within comparable families and how much arises because son-biased fertility behavior exposes girls to different family environments. A large literature shows that Indian households adjust fertility in response to the sex composition of existing children, continuing childbearing until a desired number of sons is achieved (Jensen, 2003; Barcellos et al., 2014; Jayachandran, 2017; Anukriti et al., 2022). As a result, girls are disproportionately likely to grow up in larger families and to face greater sibling competition for resources (Yamaguchi, 1989). Using an Oaxaca-style decomposition based on quasi-random variation in firstborn sex, we find that differences in household composition account for less than 10 percent of the overall gender gap in child poverty. The majority of the disparity instead arises within comparable household

types, consistent with girls receiving systematically smaller shares of household resources than boys.

Our estimated poverty measures also capture economically meaningful differences in children’s well-being. Even among children living in households with similar per-capita resources, those classified as individually poor receive lower educational investments and, in several specifications, exhibit poorer nutritional and literacy outcomes. These results indicate that the hidden poverty documented in this paper is associated with tangible differences in children’s health and human-capital accumulation. More broadly, they suggest that intrahousehold consumption allocation matters not only for measuring current deprivation, but also for the material conditions under which children develop their skills.

Finally, we connect gender poverty gaps to the literature on missing women by combining our estimates of excess girl poverty by state and age with administrative mortality data from India’s Sample Registration System (SRS). Following the framework of [Anderson and Ray \(2010, 2012\)](#), we construct measures of excess female death rates by comparing observed female mortality to benchmark mortality rates implied by male mortality and biologically plausible female survival advantages. We then ask whether states and ages with larger gender gaps in child poverty also exhibit larger excess female child mortality.

We find a strong positive relationship between excess poverty and excess mortality among girls, even when accounting for unobserved sources of heterogeneity across states and ages. A counterfactual exercise suggests that the hidden poverty gaps documented in our analysis could account for one third of excess female deaths in childhood. While this exercise is intentionally simple, it highlights a plausible micro-level mechanism linking unequal intrahousehold allocation to broader demographic outcomes.

Relation to Previous Work. Why have these disparities been difficult to document? Most household surveys do not collect comprehensive information on individual consumption. As a result, while there is abundant evidence on gender differences in specific investments and outcomes, much less is known about children’s overall command over household resources. Measures of nutrition, health, and educational attainment capture important dimensions of well-being, but they reflect the cumulative effects of many household decisions and therefore provide only indirect evidence on underlying consumption allocation.

Early empirical work attempted to infer gender bias in consumption from household demand patterns ([Deaton, 1989](#); [Subramanian and Deaton, 1991](#); [Case and Deaton, 2003](#)). The intuition is that if households favor boys, the presence of sons should systematically affect expenditure shares on observable goods. While influential, these approaches do not recover how total household resources are actually divided among children. As [Deaton \(1989\)](#) emphasized, the fundamental obstacle is the absence of individual consumption data.

More recent work addresses this limitation using collective household models ([Chiappori, 1988, 1992](#)). A key methodological advance by [Dunbar et al. \(2013\)](#) shows that, when privately consumed assignable goods such as clothing are observed, intrahousehold resource shares can be recovered from Engel curves under restrictions on preferences. These methods have since been used to measure intra-

household inequality and individual poverty among women, men, and children in a range of settings.²

Despite these advances, evidence on gender gaps among children remains limited and context-dependent. In South Asia, [Brown et al. \(2021\)](#) find modest or no gender differences in children’s allocations in Bangladesh, while work in India has focused primarily on intrahousehold disparities among adults ([Calvi, 2020](#); [Calvi and Keskar, 2021](#); [Bandyopadhyay and Maity, 2026](#)). For India specifically, there is no direct evidence quantifying differences in overall consumption allocations between boys and girls within households. Nor do we know much about how such differences translate into poverty rates.

This paper makes three main contributions. First, we provide direct estimates of gender differences in children’s command over household resources and individual poverty in rural India. Second, we quantify the role of son-biased fertility behavior in generating the observed poverty gap, distinguishing unequal allocation within comparable families from differences in the household environments in which boys and girls live. We also show that the resulting measures of individual poverty are associated with meaningful dimensions of children’s well-being. Third, we connect the literatures on intrahousehold allocation, poverty, and missing women by relating gender gaps in child poverty to excess female child mortality. In doing so, we extend the insights of [Calvi \(2020\)](#) from adulthood to childhood.³

Our findings have implications for the measurement and targeting of poverty. Policies that rely exclusively on household-level indicators implicitly assume that resources are shared equally among household members and may therefore overlook substantial deprivation among girls. In our data, many girls classified as non-poor under conventional household measures are individually poor once intrahousehold allocation is taken into account. More broadly, our results suggest that intrahousehold consumption inequality is one channel through which son preference may translate into disparities in health, human-capital accumulation, and survival. While previous research has documented such disparities and often interpreted them as reflecting differential treatment of boys and girls, direct evidence on the allocation of household consumption remains limited. By recovering individual consumption and poverty measures, our analysis makes this implicit channel explicit, and links it to both child well-being and excess female mortality in childhood.

The remainder of the paper is organized as follows. Section 2 presents the theoretical framework, describes the identification and estimation of resource shares, and outlines the construction of individual-level poverty measures. Section 3 reports the resource-share estimates and documents hidden gender gaps in child poverty. Section 4 examines the role of son-biased fertility behavior. Section 5 studies the relationship between individual poverty and children’s health and human-capital outcomes. Section 6 relates gender gaps in child poverty to missing girls. Section 7 concludes.

²See, e.g., [Dunbar et al. \(2013\)](#); [Bargain et al. \(2014\)](#); [Calvi \(2020\)](#); [Penglase \(2021\)](#); [Brown et al. \(2021\)](#); [Sokullu and Valente \(2022\)](#); [Beltramo et al. \(2023\)](#); [Calvi et al. \(2023\)](#); [Lechene et al. \(2022\)](#); [Hernandez-de Benito \(2022\)](#); [Casco \(2023\)](#); [van der Merwe \(2026\)](#). Of particular note is the recent contribution of [Aminjonov et al. \(2026\)](#), who estimate individual consumption and poverty rates for children, women, and men in 45 low- and middle-income countries using harmonized household survey data. Their findings underscore the importance of moving beyond household-level poverty measures.

³[Calvi \(2020\)](#) shows that unequal resource allocation within households contributes to the phenomenon of missing women at post-reproductive ages in India.

2 Poverty Measurement with Intra-household Inequality

Poverty is fundamentally an individual-level concept, yet it is typically measured at the household level. Standard measures classify households as poor or non-poor based on aggregate resources and household size, implicitly assuming equal sharing among members. A large literature shows that this assumption can be misleading: substantial inequality in consumption and nutrition may exist within households, so that individual living standards can differ markedly from those implied by household averages (Haddad and Kanbur, 1990; Behrman, 1988; Pitt et al., 1990; Haddad et al., 1997; D’Souza and Tandon, 2019; Brown et al., 2021, 2022).

This issue is particularly important for child poverty. If boys and girls receive different shares of household resources, children living in the same household may face different levels of deprivation, and some may be poor even when the household itself is classified as non-poor.

Because comprehensive individual consumption is rarely observed, a growing literature uses collective household models to recover individual resource shares from household expenditure data. Following Dunbar et al. (2013), these methods have been applied widely to study intrahousehold inequality and individual poverty across a range of settings and demographic groups (see Section 1 for references). Their use to study gender differences in overall consumption among children, however, remains limited.

We adopt this approach using assignable expenditures to identify resource shares for adult men, adult women, boys, and girls. Combining these shares with total household expenditure allows us to construct individual consumption and poverty measures. In this section, we describe the theoretical framework, state the assumptions required for identification, and present the empirical specification. Appendix A provides the full model and additional details.

2.1 Theoretical Framework

We model intrahousehold allocation using the collective household framework, in which household members may have distinct preferences but household allocations are Pareto efficient (Chiappori, 1988, 1992). A useful implication of this framework is that the household allocation can be represented in two stages. First, total household resources are divided across household members through *resource shares*. Second, conditional on these shares, consumption choices can be represented as if each member faced an individual shadow budget (Browning et al., 2013).

Consider a household composed of adult men and adult women, boys, and girls, indexed by $j \in \{m, w, b, g\}$. Let N_j denote the number of members of type j , and let y denote total household expenditure. The household allocates resources across member types according to

$$\max_{\{c_j\}_{j \in \{m, w, b, g\}}} \sum_{j \in \{m, w, b, g\}} N_j \omega_j u_j(c_j) \quad \text{s.t.} \quad \sum_{j \in \{m, w, b, g\}} N_j c_j = y,$$

where $u_j(\cdot)$ denotes type-specific utility and ω_j captures the Pareto weight attached to members of type j . We summarize the resulting intrahousehold consumption allocation using per-capita resource shares

η_j , defined by

$$c_j = \eta_j y, \quad \sum_{j \in \{m, w, b, g\}} N_j \eta_j = 1.$$

Thus, η_j represents the fraction of total household expenditure allocated to each individual of type j , while $N_j \eta_j$ is the total share allocated to all members of that type. Under the collective model, resource shares provide a reduced-form summary of the household allocation outcome. As shown by [Browning et al. \(2013\)](#), for a given specification of preferences and consumption technology, there exists a one-to-one relationship between resource shares and the Pareto weights that govern intrahousehold decision-making.⁴

Pareto weights are commonly interpreted as reduced-form measures of bargaining power within the household. This interpretation, however, does not imply that children’s resource shares reflect children’s own bargaining power. The collective representation associates the consumption allocated to each child type with a child-specific subutility, but this subutility need not represent a child directly participating in household decision-making. As emphasized by [Dunbar et al. \(2013\)](#), it may instead represent how parents value the consumption and well-being of their children.⁵

Our object of interest is the girl–boy gap in resource shares, $\eta_g - \eta_b$. A negative gap indicates that each girl receives a smaller per-child share of total household resources than each boy—that is, son preference is reflected not only in gaps in specific inputs and human capital outcomes, but also in the everyday allocation of consumption within the household. A gap of zero is consistent with equal per-child treatment, while a positive gap would indicate a more favorable allocation toward girls. We test whether $\eta_g - \eta_b < 0$ systematically across households and examine how the magnitude of the gap varies with household structure and characteristics. The resource share η_j captures the fraction of total household expenditure effectively commanded by a member of type j , encompassing both privately consumed goods and that member’s implicit share of household public goods. As such, it reflects the outcome of intrahousehold bargaining and depends on preferences, bargaining power, and the institutional and social environment in which households operate.

To recover resource shares η_j , we exploit *private assignable goods*—goods that are privately consumed and for which expenditures can be observed separately by recipient type. Assignable goods play a crucial role in identification because they provide a direct link between individual consumption choices and the shadow budgets determined by the intrahousehold allocation rule. In the collective model, each

⁴More generally, the collective framework can be extended to allow individual-specific preferences, Pareto weights, and resource shares. In principle, each household member may receive a distinct allocation. However, the expenditure data used here identify allocations only at the level of demographic groups. We therefore parameterize preferences and resource shares by member type—adult men, adult women, boys, and girls—and implicitly assume equal treatment of individuals within a given type. In estimation, the implied resource shares vary systematically with household composition and other observed household characteristics, allowing for substantial heterogeneity in allocations across households.

⁵For example, suppose that only the parents participate in household decision-making and that each parent’s utility is additively separable in own consumption and children’s well-being. If both parents evaluate the consumption of a given child type through a common child-specific subutility, but attach different weights to sons and daughters, adult bargaining induces an effective welfare weight for each child type. Differences in boys’ and girls’ resource shares may therefore reflect parental preferences over sons and daughters, adult bargaining power, or both, without requiring children themselves to possess bargaining power.

individual allocates the resources available to them across goods according to their preferences. When spending on a good can be attributed to a specific individual or member type, variation in that spending reflects both the individual’s preferences and the amount of resources allocated to them. However, observing assignable goods alone is not sufficient to identify resource shares without additional structure. Intuitively, higher expenditure on an assignable good for a given individual could reflect either stronger preferences for that good or a larger share of household resources. As discussed below, to disentangle these two forces, we impose mild restrictions on preferences that discipline how demand responds to changes in total resources.

In line with the previous literature, we use clothing as the assignable good. Clothing is particularly well-suited for this purpose because it is largely privately consumed, and expenditures can typically be attributed to specific individuals within the household. Moreover, clothing purchases exhibit substantial variation across individuals and respond systematically to changes in total household resources. As emphasized by [Bargain et al. \(2018\)](#), these features make clothing expenditures especially useful for estimating individual Engel curves while minimizing concerns that the good is jointly consumed or difficult to assign to specific household members.⁶ The REDS 2006 survey reports clothing and footwear expenditures separately for adult men, adult women, boys under 16, and girls under 16, allowing us to estimate type-specific Engel curves and identify four separate resource shares.

2.2 Identification

Resource shares are identified from the Engel curves of assignable clothing. Let W_{jh} denote the fraction of total household expenditure devoted to clothing for member type j in household h . An Engel curve describes how this budget share changes with total household expenditure, holding prices and other observed household characteristics fixed.

The key insight of [Dunbar et al. \(2013\)](#) is that, when an assignable good is observed separately for each member type, differences in how type-specific assignable expenditure responds to total household expenditure contain information about the allocation of overall resources within the household. Critically, *clothing budget shares* and *resource shares* are distinct objects. The level of clothing expenditure reflects both the resources allocated to a member and that member’s preferences for clothing relative to other goods. Girls may therefore spend less on clothing than boys even if their total resource shares are identical, and equal clothing spending does not imply equal resource shares. Identification comes from the response of assignable-good demand to total household resources, rather than from differences in expenditure levels.

Identification requires three conditions. First, at least one private assignable good must be observed separately for each member type. This condition is satisfied because REDS reports clothing expenditures

⁶Using unusually detailed expenditure data from Bangladesh that record private consumption at the individual level, [Bargain et al. \(2018\)](#) directly validate the collective-model approach by comparing predicted resource shares with observed intrahousehold allocations. They find that the model predicts sharing between parents and children reasonably well. Among the alternative assignable goods they consider, clothing provides the best fit and performs best in supporting the restriction used for identification proposed by [Dunbar et al. \(2013\)](#).

separately for adult men, adult women, boys, and girls.

Second, conditional on observed household characteristics, resource shares must not vary with total household expenditure over the range used for identification.⁷

Third, preferences over the assignable good must satisfy one of the semiparametric restrictions proposed by [Dunbar et al. \(2013\)](#). Our baseline imposes the Similar Across People (SAP) restriction: the slope of the individual-level clothing Engel curve with respect to the logarithm of the individual’s shadow budget is common across adult men, adult women, boys, and girls. This restriction allows the *levels* of clothing demand to differ freely across member types through type-specific intercepts, while requiring the marginal response of clothing demand to own resources to be common across types. Under SAP, differences in household-level Engel-curve slopes can therefore be attributed to differences in resource shares rather than to type-specific slope parameters.

This restriction applies only to preferences for the assignable good and does not require identical preferences across household members more generally. Appendix A examines its empirical relevance using the alternative Similar Across Types specification of [Dunbar et al. \(2013\)](#), which allows the Engel-curve slope function to differ across adult men, adult women, boys, and girls. The resulting resource-share estimates are quantitatively similar to the baseline estimates, and we cannot reject the SAP restriction.⁸

2.3 Estimation

We impose Piglog (price-independent generalized logarithmic) preferences, under which the Engel curve for the assignable good is linear in the logarithm of the individual’s shadow budget. For household h and member type $j \in \{m, w, b, g\}$, the resulting household-level clothing equation is

$$W_{jh} = N_{jh}\eta_{jh}(\alpha_{jh} + \beta_h \ln \eta_{jh} + \beta_h \ln y_h) + \varepsilon_{jh}, \quad (1)$$

where α_{jh} captures type-specific clothing preferences, β_h is the Engel-curve slope parameter that is common across member types under the SAP restriction, and ε_{jh} captures unobserved determinants of the type-specific clothing budget share.

⁷Resource shares may still depend on household characteristics that are correlated with expenditure, including demographic composition ([Dunbar et al., 2013](#)). Evidence from several settings supports the expenditure-independence restriction: [Menon et al. \(2012\)](#) find little expenditure dependence in children’s resource shares for Italian households, and [Bargain et al. \(2018\)](#) obtain similar results using data from Bangladesh.

⁸Identification also weakens when assignable-good Engel curves are flat or close to flat ([Tommasi and Wolf, 2016](#)). Appendix Table A2 reports the estimated clothing Engel-curve slopes separately for adult men, adult women, boys, and girls. All four are negative and statistically different from zero. Appendix Table A1 reports the comparison between the SAP and alternative specifications.

For a household containing all four member types, equation (1) generates the four-equation system

$$\begin{cases} W_{mh} = N_{mh}\eta_{mh}(\alpha_{mh} + \beta_h \ln \eta_{mh} + \beta_h \ln y_h) + \varepsilon_{mh}, \\ W_{wh} = N_{wh}\eta_{wh}(\alpha_{wh} + \beta_h \ln \eta_{wh} + \beta_h \ln y_h) + \varepsilon_{wh}, \\ W_{bh} = N_{bh}\eta_{bh}(\alpha_{bh} + \beta_h \ln \eta_{bh} + \beta_h \ln y_h) + \varepsilon_{bh}, \\ W_{gh} = N_{gh}\eta_{gh}(\alpha_{gh} + \beta_h \ln \eta_{gh} + \beta_h \ln y_h) + \varepsilon_{gh}. \end{cases} \quad (2)$$

Conditional on household characteristics, the slope of the type- j equation with respect to $\ln y_h$ is $N_{jh}\eta_{jh}\beta_h$. Because β_h is common across members, relative household-level Engel-curve slopes identify relative group resource shares $N_{jh}\eta_{jh}$. The adding-up condition, $\sum_j N_{jh}\eta_{jh} = 1$, then pins down their levels and hence the per-person resource shares η_{jh} .

In households containing all four member types, the system consists of four Engel curves subject to the adding-up constraint. However, many households do not contain members of all types. For example, some households contain only boys or only girls. We retain these households in the estimation. In such cases, the corresponding equations are absent, and the model is estimated on the relevant subsystem. For instance, households with adult men, adult women, and boys but no girls contribute a system of three Engel curves, with the adding-up constraint adjusted accordingly.

We estimate the system jointly using nonlinear seemingly unrelated regression (NLSUR). The equations are estimated simultaneously because unobserved determinants of expenditure may be correlated across member types. Joint estimation also allows us to impose the common- β restriction and the household-specific adding-up condition. Under multivariate normality of the disturbance vector ε_h , iterated NLSUR corresponds to maximum likelihood estimation of the nonlinear Engel-curve system. The estimator is iterated until the parameter estimates and estimated error covariance matrix converge.⁹

We allow both preferences and resource allocation to vary systematically with observed household characteristics. Let X_h denote the vector of covariates. We parameterize the type-specific preference intercepts, the common Engel-curve slope, and the resource-share functions as linear functions of these characteristics, subject to the household-level adding-up restriction.¹⁰ The baseline covariate vector includes household demographic composition, the age structure of each member type, education of adult men and women, caste, religion, and geographic region. In particular, household composition enters through the numbers of adult men, adult women, boys, and girls, while age structure enters through the average age of each member type. The nuclear-family specification omits characteristics that are constant by construction, such as the numbers of adult men and women.

Standard errors are clustered at the village level to allow for correlation in unobserved determinants of expenditure across households within villages. Inference for functions of the estimated resource

⁹For each sample, we examine 100 starting values: starting from the original initialization, we perturb the intercepts of the women's, boys', and girls' group-share functions and the common Engel-curve slope. We then select the converged candidate with the highest reported likelihood.

¹⁰In implementation, men's group share is defined residually as $s_{mh} = 1 - s_{wh} - s_{bh} - s_{gh}$, and absent child groups receive zero shares. Appendix A gives the exact parameterization.

shares, including the average girl–boy gap, uses the estimated variance–covariance matrix of the non-linear system and the delta method. For selected downstream statistics in the poverty and decomposition analyses, we assess sampling uncertainty using a leave-one-village-out jackknife. In each replication, we omit one village, re-estimate the full resource-share model, and recompute the relevant poverty or decomposition statistic on the remaining sample. The resulting leave-one-village-out estimates are used to construct jackknife standard errors and corresponding intervals.

2.4 Data and Measurement

REDS Sample and Descriptive Statistics. The Rural Economic and Demographic Survey (REDS) is a nationally representative panel survey of rural Indian households, first fielded in 1969 and subsequently followed in 1970, 1971, 1982, 1999, and 2006. We use the 2006 round because its consumption module reports clothing expenditures separately for adult men and women aged 60 or younger, boys under 16, and girls under 16. This gender-disaggregated measurement of an assignable good is the key data requirement for separately identifying boys’ and girls’ resource shares within the collective household framework described above. Earlier REDS rounds and standard expenditure surveys in India—including the NSS Consumer Expenditure Survey used in [Calvi \(2020\)](#)—do not disaggregate children’s clothing expenditures by gender, precluding analogous estimates.

We treat recorded expenditure as referring to 2005–06 and apply poverty thresholds expressed in the prices of that period.¹¹ The consumption aggregate excludes purchases of consumer durables and their repair expenditures. Households with missing monetary values for home-produced food are retained when otherwise eligible, but those entries contribute no amount to recorded total consumption.

We restrict the REDS 2006 sample to households that (i) include at least one adult man and one adult woman, (ii) have at least one child under 16, and (iii) do not employ live-in domestic workers. We further exclude mixed child-generation households, defined as households in which children under 16 include both the head’s children and grandchildren. Grandchild-only households can remain in the all-families sample, but they do not enter subsequent analyses that require a clean parental fertility history. To remove composition outliers, we retain households with at most four adults of each gender and at most six children. Finally, we trim the top and bottom 1 percent of total expenditure and the top 1 percent of each clothing budget share and require nonmissing model covariates. To hold the estimation samples fixed, these trimming rules use expenditure and budget shares before excluding durable-repair expenses; the model is then estimated using consumption and clothing budget shares net of those expenses. After these restrictions, we exclude households containing any member older than 60, so that all resident members are covered by the four modeled demographic groups. The resulting analysis sample includes 3,025 households; Appendix Figure [A1](#) documents the sample restrictions and construction.

¹¹Although REDS is referred to as the 2006 round, most interviews took place in 2008. The clothing questionnaire explicitly refers to 2005–06, whereas the separate consumption questionnaire asks about the “past year.” The reference period is therefore ambiguous. We use 2005–06 as a common reference period, following the date specified in the clothing module, and compare recorded expenditure with poverty lines for that period regardless of interview year.

Table 1: Summary Statistics

	Mean	SD	Min	Max
<i>Panel A: Household Composition</i>				
Household size	5.22	1.57	3.00	14.00
Share adult men	0.30	0.11	0.10	0.67
Share adult women	0.28	0.10	0.10	0.67
Share boys under 16	0.24	0.16	0.00	0.71
Share girls under 16	0.18	0.16	0.00	0.75
<i>Panel B: Expenditure (\$/year, 2011 PPP)</i>				
Total household expenditure	4,624	2,225	1,307	15,079
Per-capita expenditure	918	445	173	4,072
<i>Panel C: Clothing and Footwear Budget Shares (%)</i>				
Adult men	3.2	1.6	0.0	9.6
Adult women	3.3	1.7	0.0	9.7
Boys under 16	2.1	1.6	0.0	11.2
Girls under 16	1.6	1.5	0.0	8.6

Notes: Sample includes 3,025 households, including both nuclear and extended families with no member older than 60. Consumption excludes purchases of consumer durables and repairs to consumer durables. Recorded expenditures are converted to 2011 purchasing power parity (PPP) US dollars using the fixed 2005–06 CPI-AL index (358), the 2011 index (602), and the revised rural India PPP factor (13.173).

In our main analysis, we estimate the model separately over two samples: an *all-families* sample, which includes both nuclear and extended family structures (the latter include additional adult relatives, reflecting the prevalence of extended living arrangements in rural India (Calvi, 2020)) and a *nuclear-families* subsample, which restricts attention to households consisting of a married couple living only with their children (1,410 households).

We construct budget shares for private assignable goods using households’ reported annual expenditures on clothing for adult men, adult women, boys, and girls. Total household consumption expenditure is defined as the sum of food and non-durable non-food items (including clothing and footwear), excluding purchases of consumer durables and the separately reported repairs to consumer durables. Rent remain included. Clothing budget shares and individual consumption use this same expenditure aggregate.

Table 1 reports selected descriptive statistics. The average household contains 5.2 members. Adult men and adult women account for 29.6 percent and 28.2 percent of household members, respectively, while among children under 16, boys represent 23.7 percent and girls 18.4 percent. Mean annual household expenditure is Rs. 36,227 (approximately \$4,624 in 2011 PPP), corresponding to Rs. 7,190 per capita (approximately \$918 in 2011 PPP).¹² Food accounts for 58.8 percent of total consumption

¹²PPP denotes purchasing power parity. We convert recorded expenditure to 2011 PPP US dollars by dividing by $13.173 \times (358/602)$, under the maintained 2005–06 reference-period assumption. The CPI-AL index for 2005–06 is 358 and the calendar-2011 index is 602. The factor 13.173 is the revised 2011 rural consumption PPP for India reported in Ata-

expenditure. Average assignable clothing and footwear budget shares are 3.3 percent for adult women, 3.2 percent for adult men, 2.1 percent for boys, and 1.6 percent for girls, which is in line with previous studies.¹³ Women report 3.4 years of schooling on average compared to 6.1 years for men. The median age is 33 for women and 36 for men, and the median number of children per household is 2. 88.9 percent of households identify as Hindu, and 76.4 percent belong to the Scheduled Castes/Scheduled Tribes/Other Backward Classes. Geographically, the sample is concentrated in North India (51.4 percent), followed by South (26.1 percent), West (14.3 percent), and East (8.3 percent).

Measurement and Outcomes. Using the estimated resource shares, we construct individual consumption levels and compute individual poverty rates. Our baseline poverty measure follows the World Bank's extreme poverty line of \$1.90 per person per day (2011 PPP). Because REDS is a rural sample, we use the revised 2011 rural PPP conversion factor of Rs. 13.173 per international dollar and the CPI for Agricultural Labourers to express this historical World Bank line in fixed 2005–06 prices. This yields an unscaled daily threshold of 14.88 rupees ($1.90 \times 13.173 \times 358/602$); the annual threshold is 365 times this amount.

To account for differences in consumption needs across demographic groups, we construct a poverty measure based on age- and sex-specific caloric requirements from the [Indian Council of Medical Research and National Institute of Nutrition \(2020\)](#). Specifically, we scale the unscaled daily threshold by the ratio of recommended caloric requirements for each age-sex group relative to an adult reference individual. This adjustment recognizes that equal levels of consumption need not imply equal consumption adequacy across children of different ages and genders. Appendix Table A3 reports the age- and sex-specific rescaled thresholds. For comparison, we also construct an OECD equivalence-scale-adjusted poverty measure commonly used in the poverty literature: following [Dunbar et al. \(2013\)](#) and [Calvi \(2020\)](#), we assign children a poverty line equal to 60 percent of the adult threshold. Unlike the calorie-adjusted measure, this approach does not distinguish children by age or gender. Finally, we report results using the Planning Commission of India's rural poverty line of Rs. 13 per person per day in 1999 prices, adjusted to fixed 2005–06 prices.

To assess whether the individual poverty measures recovered from the collective model capture meaningful differences in children's well-being, we also examine a range of child outcomes, including nutritional status, educational investments, and learning outcomes. The REDS survey contains anthropometric information for all household members as well as detailed data on children's schooling expenditures and literacy skills. We use these outcomes to evaluate whether children classified as individually poor exhibit systematically worse outcomes than otherwise similar children living in households

manov et al. (2020, Table A.2). Using the rural PPP follows the World Bank Poverty and Inequality Platform methodology, which computes separate rural and urban PPPs for India and converts household welfare across years using consumer price indices ([World Bank, 2022](#)). We use all-India Consumer Price Index for Agricultural Labourers (CPI-AL) values from India's Labour Bureau ([Labour Bureau, Ministry of Labour and Employment, Government of India, 2012](#)).

¹³Recall that resource shares are identified from the marginal responses of type-specific clothing budget shares to total household expenditure, rather than from the levels of clothing expenditure themselves. Thus, the fact that clothing represents a relatively small share of the household budget is not by itself problematic for identification, provided the Engel curves exhibit sufficient variation.

with comparable overall resources.¹⁴ If the estimated resource shares capture genuine differences in children’s access to household resources, individual poverty should be associated with poorer health status and also with lower educational investments and weaker learning outcomes, even after conditioning on conventional household poverty measures.

3 Individual Consumption and Poverty Among Children

We now present the estimated resource shares and the implied rates of child poverty by gender. We then quantify the extent of hidden poverty generated by intra-household inequality. The next section decomposes the gender gap in child poverty into the contributions of differential resource allocation and son-biased fertility behavior.

3.1 Resource Share Estimates

Our estimates indicate large and systematic within-household gender disparities in consumption allocation among children. As shown in Table 2, boys receive a larger share of household resources than girls in both samples, on average.¹⁵ Because resource shares must sum to one, both the levels of the shares and the magnitude of the percentage-point gaps are mechanically affected by household composition: children account for a larger fraction of total household resources in smaller, nuclear households than in extended-family settings. Nevertheless, the estimates consistently indicate that boys receive substantially more resources than girls across household structures. In nuclear families, the estimated resource share of boys is 17.2 percent, compared to 10.2 percent for girls, implying a gap of -7.0 percentage points ($p < 0.001$). In the all-families sample, the corresponding average shares are 15.5 percent for boys and 11.7 percent for girls, implying a smaller gap of -3.8 percentage points ($p = 0.036$).

These gaps are economically meaningful. Individual consumption is proportional to per-person resource shares, obtained by dividing each group share by its number of members. A lower per-person share for girls implies lower individual consumption holding total household resources fixed. This within-household disparity, therefore, constitutes a direct channel through which gender differences in poverty and welfare can arise, even among children living in the same household.

¹⁴The REDS 2006 survey collected height measurements for all household members, including children aged 0–15. However, these data contain some measurement error. Heights were recorded using a feet-and-inches format that appears to have been coded inconsistently across enumerators: some entries follow a feet-inches convention (e.g., 5.06 for 5 feet 6 inches), while others appear to record decimal feet (e.g., 5.50 for 5.5 feet). After converting these measures to centimeters and applying age-specific plausibility bounds, some observations fell outside expected biological ranges. For these reasons, we compute height-for-age z-scores using internal normalization—standardizing height within age-by-sex cells using the sample mean and standard deviation rather than external WHO growth standards.

¹⁵Recall that resource shares are parameterized as functions of household characteristics and jointly estimated within the nonlinear SUR system. Table A4 in the Appendix shows that estimated shares vary systematically with household demographic composition, particularly the number and age of boys and girls. Additional boys are associated with larger boys’ group shares and smaller girls’ shares, while additional girls are associated with larger girls’ group shares and smaller boys’ shares; the strength and statistical significance of these associations vary across samples. Older children also tend to receive larger group shares within their own gender group. By contrast, parental education, caste, and religion exhibit relatively weak associations with estimated allocations once household composition is controlled for.

Table 2: Resource Share Estimates

	All Families			Nuclear Families		
	Mean	Median	SD	Mean	Median	SD
Men (η_m)	0.344	0.361	0.126	0.390	0.385	0.085
Women (η_w)	0.225	0.263	0.073	0.302	0.301	0.038
Boys (η_b)	0.155	0.154	0.048	0.172	0.166	0.040
Girls (η_g)	0.117	0.113	0.037	0.102	0.106	0.038
Girl-Boy Gap ($100 \times (\eta_g - \eta_b)$)		-3.8			-7.0	
p -value for gap = 0		0.036			<0.001	
Observations		3,025			1,410	

Notes: Nonlinear SUR estimates. All families include nuclear and extended households; nuclear families include only parents and their children. Rows report means, medians, and standard deviations of per-person shares, obtained by dividing each household's group allocation by its number of members of that type. Households containing that type receive equal weight. The girl-boy gap is the difference between these conditional mean per-child shares, in percentage points. Its two-sided p -value uses the delta method and the village-clustered covariance matrix of the fitted coefficients, treating household covariates and counts as fixed.

Figure A2 in the Appendix shows that these differences are not driven by outliers or isolated segments of the distribution. The cumulative distribution function of girls' resource shares lies uniformly to the left of that of boys in both samples. Equivalently, for any given level of resource share, girls are more likely than boys to fall below that threshold. A Kolmogorov–Smirnov test rejects equality of the two distributions in both samples ($p < 0.01$), consistent with first-order stochastic dominance of boys' allocations over girls' allocations (Table A5 in the Appendix). This pattern indicates that the gender gap in intra-household resource allocation is pervasive rather than concentrated among a small subset of households. At the same time, the distributions exhibit substantial dispersion, pointing to meaningful heterogeneity across households in the extent of intra-household inequality.

Adult allocations are also unequal. Men receive substantially larger resource shares than women—39.0 percent versus 30.2 percent in nuclear families and 34.4 percent versus 22.5 percent in the all-families sample—consistent with prior evidence on intra-household gender inequality in India (Calvi, 2020; Calvi and Keskar, 2021; Bandyopadhyay and Maity, 2026; Kapoor and Ravi, 2025).

Robustness Checks and Additional Results. We conduct several exercises to assess whether the estimated gender gap is driven by household composition and estimation sample. First, the gap remains negative when we restrict the estimation sample to households containing both boys and girls (Appendix Table A6) and when we examine households containing exactly one boy and one girl (Appendix Table A7). In the latter group, where the number of boys and girls is held fixed by construction, the predicted girl-boy gap remains sizeable. Comparisons among one- and two-child same-sex families lead to the same qualitative conclusion: predicted per-child allocations are larger for boys than for girls at a given family size (Appendix Table A8).

A complementary exercise evaluates the estimated model at a common household profile rather than

averaging boys' and girls' shares over households with different characteristics. We consider a household with one man, one woman, one boy, and one girl, holding all other covariates at their sample medians. At this common profile, girls' estimated resource shares are 7.6 percentage points lower than boys' in the all-family specification and 8.2 percentage points lower in the nuclear-family specification; both differences are statistically significant (Appendix Table A9). Because boys' and girls' shares are evaluated at exactly the same observable household characteristics, this exercise shows that the average gender gap is not simply a consequence of boys and girls being observed in different types of households.

The results are also robust to alternative specifications and sample restrictions. Because many of the socioeconomic covariates in the baseline resource-share specification are individually imprecisely estimated (see Table A4), we assess whether the main results depend on this relatively rich set of controls. We therefore estimate a substantially more parsimonious specification in which resource shares depend primarily on household demographic composition, children's ages, and region of residence. The resulting resource shares are very similar to those in the baseline model (Table A10).

We also address two potential concerns specific to the use of clothing as the assignable good. First, because clothing may be more easily shared among older children and adults, we re-estimate the model after excluding households with children aged 13–15. The estimated girl–boy gap becomes larger rather than smaller (Table A11). Second, school uniforms could generate gender differences in clothing expenditures if boys and girls attend different types of schools. Excluding households with children attending private school leaves the estimated gaps essentially unchanged (Table A12).

Finally, the direction of the gender gap is robust to alternative definitions of the estimation sample and household structure. Re-estimating the model under narrower and broader relationship restrictions continues to produce lower resource shares for girls, although the magnitude and precision of the estimates vary across samples (Table A13). We also explicitly include household members over age 60 as a fifth demographic group. The estimated girl–boy gap remains negative in this larger sample (Appendix Table A14).

Table A15 in the Appendix provides a broader perspective on the economic importance of these gender disparities by decomposing overall inequality in child consumption into between-household and within-household components using the generalized entropy measure $GE(0)$, or mean log deviation. As expected, most inequality in children's consumption arises across households. Nevertheless, intrahousehold differences account for a meaningful share of total inequality (10.4 percent in the full sample and percent among households containing both boys and girls).¹⁶

Taken together, these results show that unequal intrahousehold allocation is an important source

¹⁶These within-household estimates should be interpreted as conservative lower bounds. Our framework identifies consumption separately for boys and girls but assigns the same consumption level to children of the same gender within a household. The within-household component therefore captures gender-based differences in allocation but not inequality associated with birth order, age, or other sources of differential treatment among same-sex siblings. The fact that the within-household share rises to 17.9 percent among households containing both boys and girls is therefore particularly informative: in precisely the households in which gender-based allocation differences can be observed most directly, a nontrivial share of overall inequality in child consumption arises within rather than between families.

of the observed gender differences in child consumption. However, unequal allocation need not be the only source. Even if boys and girls were treated equally within the household, differences in the types of households in which they reside could generate additional gender disparities. In particular, if girls are disproportionately exposed to larger households, where resources are spread more thinly across children, differences in household composition may generate additional gender gaps in poverty in the aggregate. The next subsection shows how unequal allocation translates into differences in child poverty, before Section 4 decomposes the resulting poverty gap into within-household-type and household-composition components.

3.2 Hidden Gender Gaps in Child Poverty

Computing Child Poverty. Standard poverty measures assign a common poverty status to all members of a household based on per-capita expenditure, implicitly assuming equal sharing within the household. As a result, they cannot capture inequality in resource allocation within families and may systematically obscure gender disparities in welfare (Dunbar et al., 2013; Bargain et al., 2014; Brown et al., 2021).

To uncover these hidden gender gaps in child poverty, we map estimated resource shares into individual consumption. Specifically, model-implied individual consumption is computed as the product of each household member’s estimated resource share and total household expenditure. Individuals are then classified as poor if their implied consumption falls below an age-adjusted poverty threshold.

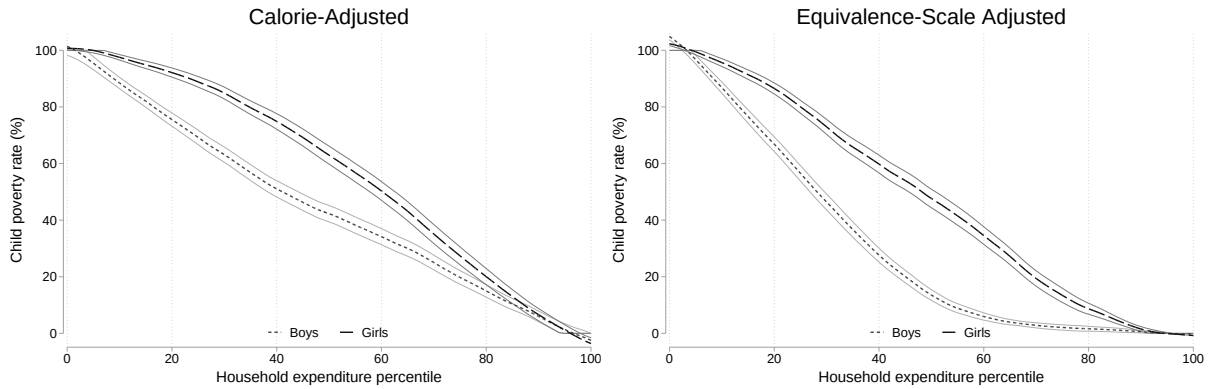
Our preferred specification uses the World Bank extreme poverty line of \$1.90 per person per day (2011 PPP), converted to an annual adult threshold in fixed 2005–06 prices (Appendix Table A3). As previously mentioned, material needs vary systematically by age and gender, however. As a result, applying a common poverty threshold to all household members may generate welfare-inconsistent comparisons. We therefore treat the World Bank poverty line as corresponding to the caloric requirements of a reference working-age adult and rescale individual poverty thresholds proportionally using age- and gender-specific caloric requirements. For comparability with earlier applications of the collective-household framework to poverty measurement, we also report results using the simpler equivalence-scale adjustment, which assigns children under age 15 a poverty threshold equal to 60 percent of the adult benchmark.

Gender Gaps in Child Poverty. Let $P_g = \Pr[\text{poor} \mid \text{girl}]$ and $P_b = \Pr[\text{poor} \mid \text{boy}]$ denote the poverty rates among girls and boys, respectively, and define the overall gender gap in child poverty as:

$$\Delta P = P_g - P_b.$$

In the full sample, 53.8 percent of girls are poor under the equivalence-scale adjustment, compared to 33.5 percent of boys, implying a gap of 20.3 percentage points. When poverty thresholds are adjusted for age- and gender-specific requirements, the estimated poverty rates increase for both groups, but substantial gender disparities remain: 63.6 percent of girls are poor compared to 48.4 percent of boys,

Figure 1: Child Poverty by Household Per-capita Expenditure



Notes: Sample: children younger than 16 in both nuclear and extended families. Within each panel, boys and girls are shown separately, with shaded 95 percent confidence bands around the poverty curves. The calorie-adjusted panel scales the adult World Bank benchmark (\$1.90/day) by Indian age-sex-specific calorific requirements from [Indian Council of Medical Research and National Institute of Nutrition \(2020\)](#). The equivalence-scale-adjusted panel assigns children under age 15 a poverty threshold equal to 60 percent of the adult World Bank benchmark. At a given expenditure percentile, a higher girls' curve indicates greater poverty among girls.

implying a gap of 15.2 percentage points. The corresponding poverty gaps are even larger among nuclear families, consistent with the stronger disparities in resource shares documented above.¹⁷

These differences are mechanically linked to the model estimates. Since individual consumption is proportional to resource shares, lower shares for girls imply lower consumption relative to boys facing the same household resource constraint. When household resources are sufficiently limited, these differences in implied consumption translate into systematically higher poverty rates for girls (as shown in Table A16 in the Appendix).

Figure 1 shows that gender gaps in poverty can persist across the household expenditure distribution, including among relatively well-off households. Figure A3 in the Appendix shows that these disparities emerge early in life and persist across most of childhood and early adolescence, although the calorie-adjusted gap narrows to approximately zero by age 15. The findings are also not specific to the World Bank poverty benchmark: using the Planning Commission's rural poverty line yields a 21.0 percentage-point girl–boy poverty gap (Appendix Table A16).

Hidden Poverty and Inequality. The *hidden* nature of these disparities becomes apparent when comparing individual poverty to standard household-level classifications. Under conventional approaches, households are classified as poor or non-poor based on per-capita consumption, and all members are assigned the same poverty status. By construction, this procedure ignores differences in resource allocation within families and can therefore generate two distinct forms of mismeasurement.

¹⁷As a benchmark, the conventional per-capita poverty rate in our all-families REDS sample is 39.5 percent using an approximately comparable 2005–06 rural poverty line, close to the 41.8 percent rural headcount reported from the 2004–05 NSS.

First, poverty may be hidden *within households*: some children may be individually poor even when their household is classified as non-poor. This phenomenon has been documented in several settings, including South Asia (Calvi, 2020; Brown et al., 2021). Second, poverty may be hidden *across genders*: because girls systematically receive smaller shares of household resources than boys, they are disproportionately likely to experience deprivation even when living in the same household.

Figure 2 reports individual poverty rates separately by child sex, household poverty status, and poverty-line adjustment. In every comparison, girls exhibit higher poverty rates than boys. Under the equivalence-scale adjustment, nearly 28.9 percent of girls living in households classified as non-poor are nevertheless individually poor, compared to 8.5 percent of boys. When poverty thresholds are adjusted for age- and gender-specific caloric requirements, the corresponding rates rise to 42.3 percent for girls and 28.8 percent for boys.

These findings imply that a substantial share of children—particularly girls—experience deprivation that is invisible to standard household-level poverty measures. Because anti-poverty programs are typically targeted using household rather than individual indicators, such children may be overlooked despite facing material deprivation.

Note that child poverty rates are not necessarily equal to 100 percent even among households classified as poor. First, unequal intrahousehold allocation implies that some children receive larger shares of household resources than others. Second, children are evaluated against age- and sex-adjusted thresholds that can differ from the unadjusted household benchmark. As a result, some children living in poor households may nevertheless have consumption levels above their individual poverty line.

Taken together, these results indicate that gender disparities in child poverty are both substantial and systematically understated by conventional household-based measures. They also motivate the next question: to what extent do these disparities reflect unequal allocation within households, as opposed to differences in the types of households in which girls and boys live? We address this question in the next section.

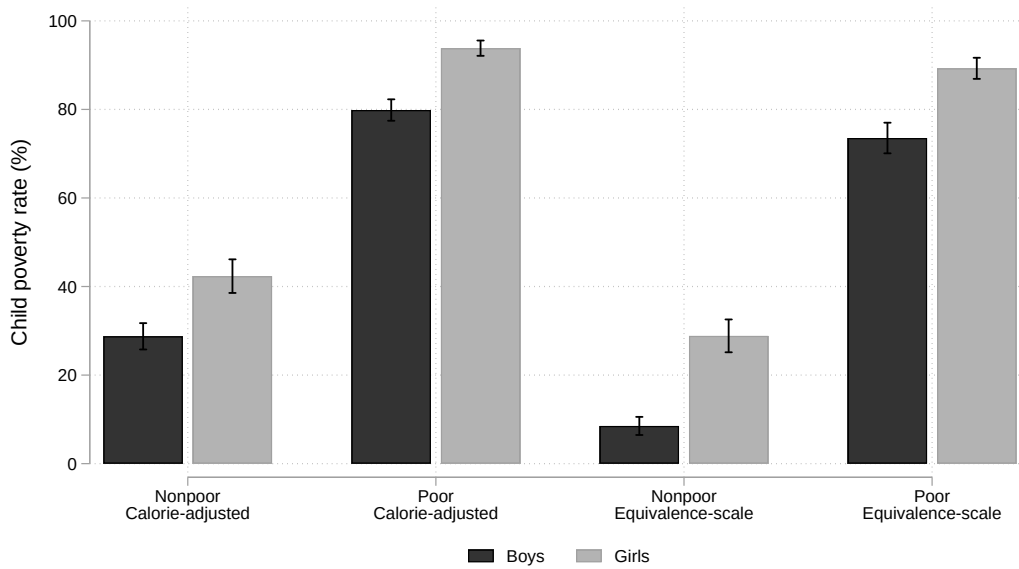
4 Son-Biased Fertility and Gender Poverty Gaps

The preceding results establish two central facts. First, girls experience higher rates of individual poverty than boys. Second, these disparities are largely invisible in standard household-level poverty measures because they arise within households rather than across households. A natural next question is, therefore, what mechanisms generate these gender gaps in child poverty?

Unequal allocation within households is one obvious candidate mechanism. Our estimates indicate that girls receive systematically smaller per-child resource shares than boys, implying lower individual consumption holding household resources fixed. However, in settings characterized by strong son preference, fertility behavior may also shape the household environments in which girls and boys grow up.

A large literature documents that parents in India adjust fertility dynamically in response to the

Figure 2: Child Poverty by Gender and Household Poverty Status



Notes: Sample: children younger than 16 in all families. Bars show individual poverty rates by sex and household poverty status. Whiskers show the original-sample estimate plus or minus 1.96 village-jackknife standard errors. Household poverty uses per-capita expenditure and the adult World Bank benchmark (\$1.90/day). Calorie-adjusted thresholds scale this benchmark by age- and sex-specific calorific requirements from [Indian Council of Medical Research and National Institute of Nutrition \(2020\)](#). Equivalence-scale-adjusted thresholds assign children under age 15 a poverty line equal to 60 percent of the adult benchmark.

sex composition of existing children, continuing childbearing until a desired number of sons is achieved (e.g., [Jensen, 2003](#); [Barcellos et al., 2014](#); [Jayachandran and Kuziemko, 2011](#); [Anukriti et al., 2022](#)). As a result, girls are disproportionately represented in larger families and therefore face greater competition for household resources. This mechanism can generate gender differences in poverty even in the absence of consumption discrimination within households.

To study the role of fertility-driven household composition, we classify households according to the sex of the firstborn child. In the cohorts represented in REDS 2006, firstborn sex is generally viewed as plausibly exogenous because sex-selective abortion at first birth remained relatively uncommon relative to later parities ([Jayachandran, 2017](#); [Anukriti et al., 2022](#)). At the same time, firstborn sex has important downstream consequences for fertility behavior under son preference. Families whose first child is a girl are more likely to continue childbearing in pursuit of a son, leading to larger completed family size, higher parity, and different sibling composition. Table A18 provides direct evidence of these fertility responses in our sample. Families whose firstborn child is a girl have significantly more children than families whose firstborn child is a boy. Moreover, among firstborn-girl households, younger siblings are substantially more likely to be boys, consistent with continued fertility in pursuit of a son.

Defining household type using firstborn sex has two advantages. First, it captures fertility responses induced by son preference while remaining plausibly orthogonal to unobserved household characteris-

tics. Second, it summarizes the cumulative consequences of dynamic fertility choices—including family size, birth order, and sibling composition—without conditioning directly on endogenous outcomes such as completed fertility or parity.¹⁸

We begin by documenting how gender gaps in consumption allocation within families vary across household composition. Figure A4 in the Appendix reports the girl–boy gap in per-child resource shares (that is, the difference between the average estimated resource shares for girls and boys) by firstborn sex, older-brother status, and the girls-to-boys ratio within the household. Several patterns emerge. First, the resource-share gap is substantially larger in households where the firstborn child is a girl than in households where the firstborn is a boy. This pattern is consistent with stronger son-biased fertility behavior in firstborn-girl households, which tend to be larger and more resource-constrained. Second, girls receive systematically smaller shares regardless of whether they have an older brother, indicating that unequal allocation is not driven solely by direct competition with older male siblings. Third, the gender gap in resource shares becomes substantially larger in households where girls outnumber boys. In households with more girls than boys, girls receive substantially smaller per-child allocations relative to boys, whereas the gap is close to zero in households where girls are relatively scarce.

Appendix Figures A5 and A6 show that these differences in allocation translate into substantial disparities in poverty. The gender gap in child poverty is considerably larger among households where the firstborn child is a girl, and increases sharply with the relative number of girls in the household. In households where girls outnumber boys, the poverty gap becomes particularly large, suggesting that the interaction between son-biased fertility stopping behavior and unequal intrahousehold allocation has major consequences for child deprivation.

To formally study the role of son-biased fertility, we perform a Oaxaca-style decomposition (Oaxaca, 1973; Blinder, 1973) of the overall gender gap in child poverty. The decomposition separates the gap into a within-type component and a household-composition component. Because the decomposition relies on fertility behavior and sibling composition, we restrict the analysis to nuclear families.¹⁹ The household-composition component captures differences in exposure to family environments generated by son-biased fertility-stopping behavior, while the within-type component captures residual gender differences in poverty among children living in comparable family environments. Let $t \in \{0, 1\}$ index household type, where $t = 1$ denotes households whose firstborn child is a girl and $t = 0$ denotes households whose firstborn child is a boy. For each child sex $s \in \{g, b\}$, define

$$\pi_{s,t} \equiv \Pr(T = t \mid S = s)$$

¹⁸For detailed evidence of the quasi-randomness of the sex of the firstborn, see Anukriti et al. (2022).

¹⁹The all-families estimation sample excludes households in which children under 16 include both the head’s children and grandchildren, but grandchild-only households can remain in broader all-families analyses. In the nuclear sample used here, under-16 children are the head’s own children, so firstborn sex and sibling composition can be mapped cleanly to a parental fertility history.

as the share of children of sex s who live in households of type t , and

$$P_{s,t} \equiv \Pr(\text{poor} = 1 \mid S = s, T = t)$$

as the poverty rates within that household type. The poverty rates among girls and boys can therefore be written as

$$P_s = \sum_{t \in \{0,1\}} \pi_{s,t} P_{s,t}.$$

We then construct a counterfactual poverty rate for girls:

$$P_g^{cf} = \sum_{t \in \{0,1\}} \pi_{b,t} P_{g,t},$$

which essentially captures what girls' poverty rate would be if girls faced the same distribution of household types as boys, while preserving the within-household poverty outcomes observed for girls.

Adding and subtracting P_g^{cf} yields the following decomposition of the gender gap in child poverty:

$$\Delta P = (P_g^{cf} - P_b) + (P_g - P_g^{cf}).$$

The first term captures the *within-type component*: differences in poverty between girls and boys holding constant household type. Because household type is defined by the sex of the firstborn child, this component reflects gender differences in poverty that arise within comparable family environments and is therefore most closely related to unequal intrahousehold allocation. The second term captures the *household composition component*: differences arising because girls and boys are differentially exposed to household types associated with son-biased fertility behavior. In particular, girls are more likely to reside in families whose firstborn child is a girl and which, due to son-biased stopping rules, tend to be larger and contain more children.

Table 3 reports the results. Girls have higher poverty rates in nuclear families than in the full sample, whereas boys have lower poverty rates, consistent with the larger gender gap in per-child resource shares in nuclear households (see Section 3). Overall, the vast majority of the gender gap in child poverty is explained by within-type differences rather than by differential household composition. Under the equivalence-scale-adjusted calculations, approximately 92.7 percent of the total gender gap in child poverty is attributable to differences in poverty between girls and boys within the same firstborn-sex household type, while only 7.3 percent is explained by composition effects. When differences in needs by age and gender are taken into account, 90.4 percent of the gap is attributable to within-type differences, with the remaining 9.6 percent explained by differential exposure to different household types associated with son-biased fertility behavior.

This conclusion is robust to accounting for sampling uncertainty in the estimated resource shares. Appendix Table A19 reports leave-one-village-out jackknife standard errors obtained by re-estimating the resource-share model, poverty rates, and decomposition in each replication. The composition com-

Table 3: Decomposition of the Girl-Boy Child Poverty Gap

	Girls (%)	Boys (%)	Total Gap (p.p.)	Within p.p. (%)	Composition p.p. (%)
Calorie-adjusted	82.0	42.6	39.3	35.5 (90.4)	3.8 (9.6)
Equivalence-Scale Adjusted	75.0	26.8	48.2	44.7 (92.7)	3.5 (7.3)

Notes: The table decomposes the nuclear-family girl-boy poverty gap into within-household-type and household-composition components. Household type is indexed by the sex of the firstborn child. The within component is the gap that would remain if girls had the same distribution of firstborn-sex household types as boys. The composition component is the residual contribution from girls' and boys' differential exposure to firstborn-girl and firstborn-boy households. Values in parentheses report each component as a percentage of the total girl-boy gap.

ponent remains small: the upper bound of its 95 percent interval is 5.55 percentage points under both poverty measures, compared with total gender gaps of 39.3 and 48.2 percentage points, respectively. So, even at the upper bound of the interval, differences in household composition account for only a modest fraction of the overall gender gap in child poverty.

5 Child Poverty and Well-Being

An important question is whether the individual poverty measures recovered from the collective model capture meaningful differences in children's well-being beyond those reflected in conventional household poverty measures. If low consumption and poverty risk matter for children's well-being, then children classified as poor based on their own estimated individual consumption should exhibit worse outcomes even after accounting for overall household resources.

To investigate this question, Table 4 examines the relationship between child poverty and a range of child outcomes spanning health, educational investments, and learning. In each specification, we include both the child's individual poverty status and a conventional household per-capita poverty indicator. This allows us to assess whether the information contained in the estimated individual consumption has predictive power beyond that provided by standard household-level measures.

The first set of results considers height-for-age z-scores (HAZ), a widely used indicator of nutritional status. Under the calorie-adjusted measure, the estimated coefficients on individual poverty are -0.083 standard deviations for girls and -0.063 for boys, conditional on household poverty status, child age, and household composition. Although these estimates are imprecisely estimated, their magnitudes are economically nontrivial: individual poverty is associated with HAZ that is about 0.08 standard deviations lower for girls and 0.06 standard deviations lower for boys. Under the equivalence-scale-adjusted measure, the corresponding estimates are larger, at 0.13 standard deviations for girls and 0.10 for boys, with the estimate for girls statistically significant at the 5 percent level.

The second set of outcomes examines educational expenditures. Under the calorie-adjusted measure, individually poor girls receive approximately 55.3 log points (about 42.5 percent) less educa-

Table 4: Individual Poverty and Child Human Capital Outcomes

	Girls				Boys			
	Health	Human Capital			Health	Human Capital		
	HAZ	Education Spending	Advanced Reading	Advanced Writing	HAZ	Education Spending	Advanced Reading	Advanced Writing
<i>Panel A. Calorie-Adjusted Poverty Measure</i>								
Individually Poor	−0.083 (0.053)	−0.553*** (0.082)	−0.142*** (0.035)	−0.115*** (0.033)	−0.063 (0.053)	−0.358*** (0.062)	−0.069** (0.030)	−0.049 (0.031)
HH Per-Capita Poor	−0.038 (0.053)	−0.342*** (0.071)	−0.017 (0.034)	−0.020 (0.031)	−0.073 (0.051)	−0.450*** (0.064)	−0.041 (0.032)	−0.057* (0.033)
Observations	2,931	1,757	1,781	1,781	3,660	2,367	2,387	2,387
<i>Panel B. Equivalence-Scale-Adjusted Poverty Measure</i>								
Individually Poor	−0.127** (0.059)	−0.335*** (0.097)	−0.010 (0.039)	−0.009 (0.035)	−0.099 (0.066)	−0.125** (0.059)	0.001 (0.033)	−0.039 (0.034)
HH Per-Capita Poor	−0.008 (0.055)	−0.395*** (0.083)	−0.075** (0.037)	−0.066* (0.034)	−0.043 (0.057)	−0.565*** (0.066)	−0.079** (0.033)	−0.059* (0.034)
Observations	2,931	1,757	1,781	1,781	3,660	2,367	2,387	2,387

Notes: The table reports associations between individual poverty and child outcomes. Height-for-age z-scores (HAZ) are internally standardized by age and sex. Education Spending is measured as $\ln(1 + \text{annual Rs.})$. Advanced Reading and Advanced Writing are indicators equal to one if the child can read or write in both English and their mother tongue. Panel A defines individual poverty using age- and sex-specific calorie-adjusted poverty thresholds. Panel B defines individual poverty using an equivalence-scale-adjusted poverty threshold, which assigns children under age 15 a poverty line equal to 60 percent of the adult World Bank benchmark. HH Per-Capita Poor is based on standard household per-capita expenditure. All specifications additionally control for child age and household composition (numbers of boys, girls, adult men, and adult women). Village-clustered standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

tional spending than otherwise similar non-poor girls, while individually poor boys receive roughly 35.8 log points (about 30.1 percent) less. These are large differences and the corresponding coefficients are precisely estimated for both girls and boys. The negative association also appears under the equivalence-scale-adjusted measure.

The final set of outcomes considers advanced literacy skills, measured by the ability to read and write in both English and the child's mother tongue. Under the calorie-adjusted measure, individual poverty is associated with a 14.2 percentage-point lower probability of advanced reading and a 11.5 percentage-point lower probability of advanced writing among girls. The reading association is statistically significant for both girls and boys, while the writing association is statistically significant only for girls.²⁰

²⁰These patterns are not driven by children attending private schools, for whom both education expenditures and clothing requirements may differ systematically. Appendix Table A20 excludes households with an enrolled child identified as attending a private school and re-estimates the resource-share model on the restricted sample. The results are very similar.

6 Gender Gaps in Child Poverty and Missing Girls

The previous sections document hidden gender disparities in child consumption and poverty. Girls receive systematically smaller resource shares than boys, experience significantly higher rates of individual poverty, and exhibit worse health and human capital outcomes conditional on household poverty. These findings naturally raise a broader question: can hidden inequalities in everyday consumption contribute to the large deficits in female survival documented in India?

Beginning with [Sen \(1990\)](#), a large literature has documented the phenomenon of *missing women*, defined as the shortfall in the number of surviving females relative to benchmark populations with similar demographic structures. Sex-selective abortion and differential investment in girls in infancy and early childhood have received substantial attention in this literature. A large body of evidence documents systematic gender disparities in health investments, nutrition, and medical care in India and other parts of South Asia ([Das Gupta, 2005](#); [Barcellos et al., 2014](#); [Oster, 2009](#); [Jayachandran and Kuziemko, 2011](#)). More recent work has also emphasized the interaction between fertility decline, son preference, and prenatal sex selection ([Jayachandran, 2017](#); [Anukriti, 2018](#); [Anukriti et al., 2022](#)). At the same time, a growing body of work shows that excess female mortality cannot be explained solely by prenatal selection or discrimination at birth. [Anderson and Ray \(2010, 2012\)](#) demonstrate that excess female mortality in India is highly age-specific and extends well beyond birth.

Our poverty estimates above imply large gender asymmetries in individual deprivation even among children living in households that would appear non-poor according to conventional household-level measures. If unequal access to household resources translates into poorer nutrition, weaker health, and lower resilience among girls, then states and age groups with larger girl-boy poverty gaps should also exhibit larger excess female mortality. To test this hypothesis, we follow [Anderson and Ray \(2010, 2012\)](#), who provide an intuitive and tractable measure of missing women by comparing observed female mortality to a benchmark female mortality rate. We then conduct a simple counterfactual exercise in the spirit of [Calvi \(2020\)](#) relating excess female mortality to the gender gap in individual poverty estimated from the REDS data. [Calvi \(2020\)](#)'s focus is on excess female mortality in adulthood, especially at post-reproductive ages. Here, instead, we focus on infancy, childhood, and early adolescence. The exercise is intentionally simple, but it provides a quantitative benchmark for assessing whether the magnitudes of the hidden poverty gaps documented in this paper are large enough to plausibly account for differences in female child survival.

To measure excess female mortality, we combine state-age poverty gaps with age-specific mortality data from the 2011 Sample Registration System (SRS). Let d_{sa}^g and d_{sa}^b denote the observed death rates per 1,000 girls and boys in state s and age group $a \in \{0-4, 5-9, 10-14\}$, respectively, and let N_{sa}^g denote the female population in that state-age cell. Following [Anderson and Ray \(2010\)](#), let r_a denote the benchmark female-to-male mortality ratio, equal to 0.80, 0.76, and 0.67 for ages 0-4, 5-9, and 10-14, respectively. These benchmark ratios capture the biologically expected female survival advantage in low-discrimination mortality environments. The counterfactual female death rate implied by the

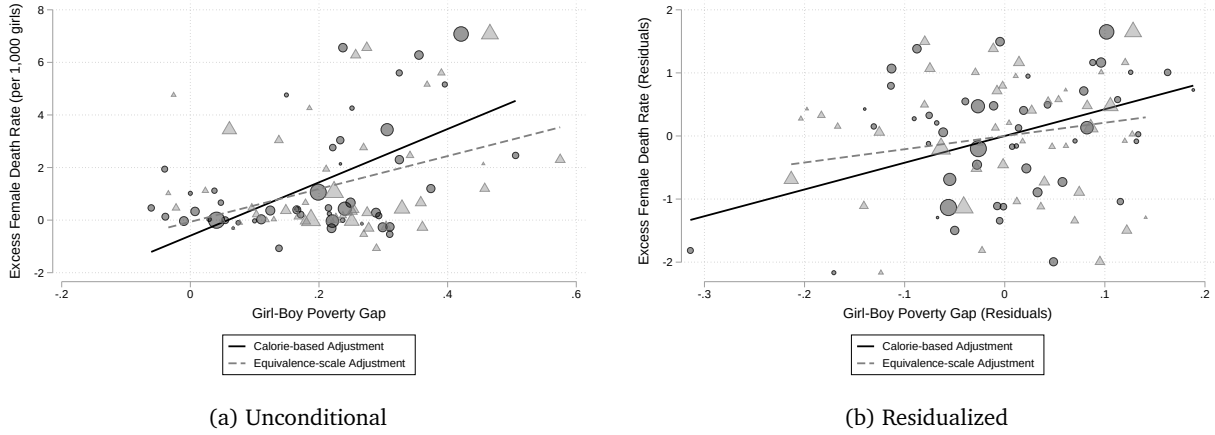


Figure 3: Gender Gaps in Child Poverty and Missing Women

Notes: Each point is a state-age cell for the 17 states common to REDS and SRS and age groups 0–4, 5–9, and 10–14. The vertical axis reports rural excess female deaths per 1,000 girls; the horizontal axis reports the girl-boy poverty gap. Circles and solid lines use calorie-adjusted poverty; triangles and dashed lines use equivalence-scale-adjusted poverty. Panel A shows the pooled relationship. Panel B removes state and age-group fixed effects from both variables. Point sizes and fitted regressions are weighted by rural female population.

benchmark is $\tilde{d}_{sa}^g = r_a d_{sa}^b$, and the excess female death rate in age group a and state s is defined as

$$EFDR_{sa} = d_{sa}^g - \tilde{d}_{sa}^g = d_{sa}^g - r_a d_{sa}^b.$$

Positive values indicate that girls die at higher rates than predicted by the benchmark female-male mortality relationship. Intuitively, $EFDR_{sa}$ measures the extent to which female mortality exceeds the level expected given observed male mortality in the same demographic environment.

For each state s and child age group a , let P_{sa}^g and P_{sa}^b denote the estimated poverty rates of girls and boys in the REDS sample, respectively, and define the girl-boy poverty gap as $\Delta P_{sa} = P_{sa}^g - P_{sa}^b$.

Figure 3 provides a graphical summary of the relationship between gender gaps in child poverty and excess female mortality. Each point represents a state-age cell formed by combining one of the 17 states that overlap between the REDS and SRS data with one of three child age groups (0–4, 5–9, and 10–14), weighted by the size of the female population in that cell. Under both poverty definitions, state-age cells with larger girl-boy poverty gaps tend to exhibit higher excess female mortality. Importantly, the positive relationship remains visible after residualizing both variables with respect to state and age-group fixed effects, indicating that the association is not driven solely by persistent differences across states or by the age profile of mortality.

Table 5 further quantifies this relationship. Across specifications, larger girl-boy poverty gaps are associated with higher excess female mortality, although the association is stronger when poverty is measured using age- and gender-specific caloric requirements. For the calorie-adjusted measure, a 10-percentage-point increase in the girl-boy poverty gap is associated with between 0.4 and 0.97 additional

Table 5: Gender Gaps in Child Poverty and Missing Girls

Measure	Marginal Effect	Counterfactual Missing Girls	Explained Share
<i>Panel A. Pooled, Population-Weighted (No Fixed Effects)</i>			
Calorie-Adjusted	9.67*** (2.09)	70,935	62.4%
Equivalence-Scale Adjusted	6.31** (2.54)	102,835	45.5%
<i>Panel B. State and Age Fixed Effects, Population-Weighted</i>			
Calorie-Adjusted	3.95* (1.98)	126,500	33.0%
Equivalence-Scale Adjusted	2.01 (1.79)	150,699	20.2%

Notes: The dependent variable is the signed excess female death rate per 1,000 girls. REDS child poverty gaps are merged to 2011 rural SRS mortality rates and rural female populations for the child age bins 0–4, 5–9, and 10–14 across the 17 states common to both datasets. All specifications are weighted by the rural female population in each state-age cell. Panel A is pooled and includes no fixed effects. Panel B additionally includes state and age-group fixed effects. The final column reports the share of estimated missing girls statistically associated with the gender gap in child poverty. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

excess female deaths per 1,000 girls, depending on the specification. As expected, the relationship is strongest in the unconditional regressions. However, it remains positive and statistically significant at the 10 percent level after controlling for state and age-group fixed effects. This is noteworthy because identification in this specification comes entirely from within-state variation across age groups.

To assess the quantitative importance of these estimates, we conduct a simple counterfactual exercise. For each state-age cell, we predict the component of excess female mortality associated with the observed gender gap in child poverty and then consider a counterfactual in which $\Delta P_{sa} = 0$. The estimated coefficients imply a reduction in the excess female death rate of $\hat{\beta} \Delta P_{sa}$ deaths per 1,000 girls. Multiplying this predicted reduction by the corresponding female population yields the implied reduction in the number of excess female deaths in state s and age group a . We then aggregate these reductions across state-age cells and compare the result to the total number of estimated missing girls in our sample. The results suggest that excess girl poverty may explain between 33.00 and 62.4 percent of missing girls in our sample. Repeating the exercise with the equivalence-scale-adjusted poverty measure yields somewhat smaller estimates, with implied explained shares ranging from 20.2 to 45.5 percent. Appendix Table A21 shows that the qualitative conclusions are robust to alternative specifications.

These estimates should be interpreted with caution. The poverty and mortality measures are drawn from different data sources, and female child mortality is influenced by many factors beyond material deprivation. At the same time, the specifications with fixed effects control for broad differences in mortality and poverty across states and stages of childhood, ensuring that the estimated relationships are not driven by persistent unobserved differences.

Even viewed as a descriptive benchmark, the results are quite striking. Hidden gender disparities in child poverty are systematically associated with excess female mortality, and the estimated relationship implies that unequal access to household resources and its implications for poverty may account for a substantial share of missing girls in childhood. More broadly, the findings suggest that intrahousehold allocation is an important channel through which son preference translates into disparities in child survival, helping to connect the literatures on intrahousehold allocation, poverty, and missing women.

7 Conclusion

This paper studies gender differences in child poverty in India through the lens of intrahousehold allocation. While a large literature has documented son preference through disparities in fertility behavior, health investments, education, and survival, much less is known about how household resources are allocated between boys and girls on a day-to-day basis. Using a collective household model and detailed expenditure data from rural India, we estimate individual consumption levels for children and construct poverty measures at the individual rather than household level.

We find substantial gender disparities in the allocation of household resources. Girls receive systematically smaller consumption shares than boys, resulting in significantly higher rates of individual poverty. A substantial fraction of poor girls reside in households that would be classified as non-poor using standard per-capita consumption measures, implying that household-based poverty statistics substantially understate both the prevalence of child poverty and the extent of gender inequality among children. Our analysis shows that most of the observed gender gap in child poverty arises within household types rather than through differential exposure to household environments generated by son-biased fertility behavior.

Combining our poverty estimates with state-by-age mortality data, we also show that areas with larger excess girl poverty exhibit higher excess girl mortality. A counterfactual exercise suggests that hidden gender gaps in child poverty can account for a substantial fraction of India's missing girls. Echoing results in [Calvi \(2020\)](#), this finding points to a potentially important link between intrahousehold consumption inequality, individual poverty, and missing women in the Indian context. More generally, our results suggest that understanding gender disparities in child well-being requires looking beyond household averages and examining how resources are distributed among children within the household.

An important question for future research is whether the patterns documented here persist in contemporary India. Recent evidence suggests that some manifestations of son preference have weakened over the past two decades, even as other dimensions of gender inequality remain persistent ([Jayachandran, 2023](#)). Yet little is known about whether these broader demographic changes have been accompanied by comparable shifts in the allocation of resources within households, or whether hidden gender gaps in child poverty have narrowed over time.

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Online Appendix

A Identification and Estimation: Details

This appendix provides formal details on the identification and estimation of intrahousehold resource shares in the collective model used in the paper. The theoretical framework and identification strategy follow [Browning et al. \(2013\)](#) and [Dunbar et al. \(2013\)](#), with the key difference that we identify separate resource shares for boys and girls. As in [Calvi \(2020\)](#), we consider a general framework that accommodates both nuclear and extended families.

A.1 Collective Framework and Resource Shares

Consider a household composed of adult men, adult women, boys under age 16, and girls under age 16, indexed by $j \in \{m, w, b, g\}$. Let N_j denote the number of household members of type j , and let y denote total household expenditure. Households differ in observable characteristics, including demographic composition, age structure, education, caste, religion, region, and family structure. These characteristics may affect both preferences and the allocation of resources within the household.

Let $h = (h^1, \dots, h^K)$ denote observed household purchases of K goods and let

$$x_j = (x_j^1, \dots, x_j^K)$$

denote the vector of private-good equivalents consumed by a representative member of type j . We allow for economies of scale in consumption through a Barten consumption technology. Specifically, there exists a $K \times K$ matrix A such that

$$h = A \sum_{j \in \{m, w, b, g\}} N_j x_j.$$

This technology transforms individual private-good equivalents into observed household purchases and permits goods to exhibit varying degrees of publicness within the household.

Members of each type derive utility from consumption according to $U_j(x_j)$, where $U_j(\cdot)$ is increasing, strictly quasi-concave, and continuously differentiable. The household is assumed to allocate resources efficiently. Consequently, observed allocations can be represented as the solution to a Pareto-weighted social welfare problem:

$$\max_{\{x_j\}} \sum_{j \in \{m, w, b, g\}} N_j \omega_j U_j(x_j)$$

subject to

$$h = A \sum_j N_j x_j, \quad p'h = y,$$

where ω_j denotes the Pareto weight attached to members of type j .

Pareto weights summarize the influence of different household members in the allocation process and are often interpreted as measures of bargaining power. However, they are not directly observable and depend on the cardinalization of utility functions. Following [Browning et al. \(2013\)](#), we therefore focus on resource shares rather than Pareto weights.

Let η_j denote the per-capita resource share allocated to members of type j . Resource shares satisfy

$$\sum_{j \in \{m, w, b, g\}} N_j \eta_j = 1,$$

and imply that each member of type j controls resources equal to $\eta_j y$. Resource shares summarize the outcome of the household allocation process and are invariant to monotonic transformations of utility. Under the collective model, and for a given specification of preferences and consumption technology, there exists a one-to-one relationship between Pareto weights and resource shares ([Browning et al., 2013](#)).

A standard result from collective models is that the household problem can be decentralized into two stages. First, total household resources are allocated across member types through the resource shares η_j . Second, conditional on receiving resources $\eta_j y$, each member type chooses its preferred consumption bundle by solving

$$\max_{x_j} U_j(x_j)$$

subject to the shadow budget constraint

$$(A'p)'x_j \leq \eta_j y.$$

The resulting indirect utility functions can be substituted into the household optimization problem, yielding a reduced-form representation in which resource shares completely summarize intrahousehold allocation. Our objective is to estimate these resource shares.

A.2 Assignable Goods and Engel Curves

Standard expenditure surveys do not observe individual consumption directly. Identification instead exploits private assignable goods: goods that are privately consumed and recorded separately by recipient type ([Dunbar et al., 2013](#)). Let W_j denote the household budget share devoted to the assignable good for type j . In our application, the assignable good is clothing and footwear, which REDS 2006 records separately for adult men, adult women, boys, and girls.

For assignable goods, household demand admits the representation

$$W_j(y, p) = N_j \eta_j w_j(A'p, \eta_j y), \tag{A1}$$

where $w_j(\cdot)$ is the shadow-budget demand function for type j , and $A'p$ denotes shadow prices allowing for economies of scale ([Browning et al., 2013](#)).

A.3 Piglog Preferences

Under Piglog preferences, Engel curves linear in $\ln y$ and resource shares are by construction independent of y . The resulting assignable-good Engel curve system is:

$$W_m = N_m \eta_m (\alpha_m + \beta_m \ln \eta_m + \beta_m \ln y), \quad (\text{A2})$$

$$W_w = N_w \eta_w (\alpha_w + \beta_w \ln \eta_w + \beta_w \ln y), \quad (\text{A3})$$

$$W_b = N_b \eta_b (\alpha_b + \beta_b \ln \eta_b + \beta_b \ln y), \quad (\text{A4})$$

$$W_g = N_g \eta_g (\alpha_g + \beta_g \ln \eta_g + \beta_g \ln y), \quad (\text{A5})$$

where α_j and β_j are preference parameters for type j .

A.4 Identification

Identification follows [Dunbar et al. \(2013\)](#) and requires:

1. at least one assignable good observed for each household member type,
2. resource shares that do not vary with total expenditure,
3. a semiparametric restriction on preferences.

The first requirement is satisfied by the REDS expenditure module, which separately records clothing, footwear, and tailoring expenditures for adult men, adult women, boys, and girls. These goods are privately consumed and can therefore be assigned to specific household member types.

The second requirement assumes that, conditional on household characteristics, resource shares are independent of total expenditure. Intuitively, increases in household resources affect individual consumption through the Engel curves rather than through systematic changes in the allocation rule itself. This assumption is standard in the collective-demand literature and is necessary to distinguish resource allocation from preference heterogeneity ([Menon et al., 2012](#); [Bargain et al., 2018](#)). This is also implied by construction by Piglog preferences.

For the third requirement, we impose the Similar Across People (SAP) restriction from [Dunbar et al. \(2013\)](#):

$$\beta_m = \beta_w = \beta_b = \beta_g \equiv \beta.$$

Under SAP, household members are allowed to differ in their overall demand for the assignable good through type-specific intercepts, but the slope of the Engel curve with respect to resources is assumed to be common across types. This restriction is substantially weaker than imposing identical preferences and is sufficient to separate preference parameters from resource shares. Under SAP, relative Engel-curve slopes identify relative resource shares, while the adding-up constraint determines their levels.

These identifying assumptions are transparent and testable with appropriate data. Previous work has tested SAP in similar settings with richer data (see [Bargain et al. \(2018\)](#) and [Brown et al. \(2021\)](#)).

in Bangladesh), finding it to be a reasonable structure to be imposed on the data. Here, we assess its empirical relevance by estimating a more flexible Similar Across Types (SAT) specification in which the Engel-curve slope function is allowed to differ across adult men, adult women, boys, and girls. Specifically, we replace the common β function with type-specific β_j , which we parameterize using the same household characteristics as in the baseline specification minus the number of adult men, women, boys, and girls. We refer the reader to [Dunbar et al. \(2013\)](#) for a detailed proof about this alternative approach to identification.

Table [A1](#) reports the results. Allowing for type-specific Engel-curve slopes has very little effect on the estimated resource shares. Under the preferred SAP specification, adult men, adult women, boys, and girls receive average resource shares of 0.344, 0.225, 0.155, and 0.117, respectively. The corresponding SAT estimates are 0.347, 0.228, 0.156, and 0.114. The implied girl–boy resource-share gap changes only modestly, from -3.8 to -4.2 percentage points. A Wald test fails to reject equality of the four slope functions.

Finally, note that identification weakens when Engel curves are close to flat because resource shares become only weakly related to observed expenditure responses ([Tommasi and Wolf, 2016](#)). In our application, estimated Engel-curve slopes are statistically different from zero for all four member types, alleviating concerns about weak identification. Appendix Table [A2](#) reports separate Engel-curve estimates for men, women, boys, and girls. The estimated slopes are all negative, statistically significant.

A.5 Estimation

We estimate the Engel-curve system using nonlinear SUR, imposing both the SAP restriction and the adding-up condition on resource shares. Estimation includes households that do not contain all four demographic groups: an absent child group’s observed budget share and fitted contribution are both set to zero.

A key feature of the specification is that resource shares and preference parameters are allowed to vary systematically across households. Rather than estimating a single set of resource shares for the entire sample, we parameterize both preferences and allocations as functions of observed household characteristics:

$$\eta_j = \mathbf{x}'\boldsymbol{\delta}_j, \quad \alpha_j = \mathbf{x}'\boldsymbol{\gamma}_j, \quad \beta = \mathbf{x}'\boldsymbol{\phi},$$

where $\mathbf{x}$ includes household composition, demographic structure, education, caste, religion, and region indicators.

This parameterization allows intrahousehold allocations to differ across households with different demographic and socioeconomic characteristics. For example, the resource shares assigned to boys and girls may vary with the number of children in the household, the presence of adult men and women, children’s and adults’ ages, parental education, caste, religion, or geographic location. Similarly, preferences for assignable clothing expenditures are allowed to vary systematically across households through the α_j parameters. As a result, the model accommodates substantial heterogeneity in both preferences

Table A1: Testing the SAP Restriction

	Men	Women	Boys	Girls	Girl–Boy Gap	Wald Test for SAP
SAP restriction	0.344	0.225	0.155	0.117	-3.8	–
SAT restriction	0.347	0.228	0.156	0.114	-4.2	$\chi^2(30) < 0.001$, $p > 0.999$

Notes: Sample: all families without members older than 60 (N=3,025). SAP denotes the baseline common-slope model. SAT allows separate slope functions for men, women, boys, and girls, with region, education, age, caste, and religion covariates but no member counts. Each resource share is the allocation to one member of the indicated type: the estimated group allocation is divided by the number of members of that type within each household. Summary statistics give equal weight to households containing that type. The village-clustered Wald test tests equality of the four SAT slope functions; the baseline SAP slope function also includes member counts. The SAT estimate remains at its common-slope starting values, so the near-zero Wald statistic is not informative about the SAP restriction.

Table A2: Engel Curve Slope Estimates

<i>Panel A. Group-specific Engel curves: all families</i>				
	Men	Women	Boys	Girls
Log expenditure	–0.0071*** (0.0011)	–0.0076*** (0.0014)	–0.0049*** (0.0010)	–0.0031*** (0.0009)
Observations	3,025	3,025	2,465	2,037
R^2	0.163	0.163	0.310	0.312
<i>Panel B. LPW identification diagnostic</i>				
	All Families			
Mean β	–0.0214 (0.0032)			
p -value for mean $\beta = 0$	< 0.001			
Households with significant β (%)	88.6			
Boys only	93.9			
Girls only	70.5			
Boys and girls	91.9			

Notes: Panel A reports regressions of each group's clothing budget share (including footwear and tailoring) on log expenditure and controls for region, education, ages, household composition, caste, and religion. The sample includes all families; boys' and girls' equations include households with at least one member of the corresponding group. Panel B reports the identification diagnostic of [Lechene et al. \(2022\)](#) for all families (N=3,025). Separate regressions are estimated for households with boys only, girls only, or both boys and girls. We regress the total assignable-clothing budget share on log household expenditure, the number of members in each group present in the household, the logarithms of these numbers, and the controls listed above. Interactions between log expenditure and member counts allow the expenditure slope to vary with household composition. Standard errors are clustered by village, treating covariates as fixed. [Lechene et al. \(2022, p. 701\)](#) suggest that household-specific expenditure slopes be statistically different from zero for at least 75% of the sample as a diagnostic benchmark for identification. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

and allocations while maintaining the restrictions required for identification.

We impose the adding-up restriction by defining men's total share residually:

$$N_m \eta_m = 1 - N_w \eta_w - N_b \eta_b - N_g \eta_g.$$

This guarantees that the estimated resource shares satisfy

$$\sum_j N_j \eta_j = 1$$

for every household in the sample.

B Additional Figures

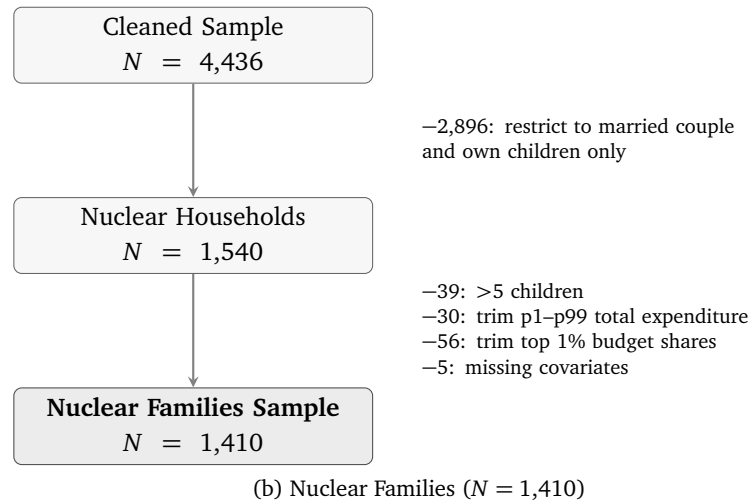
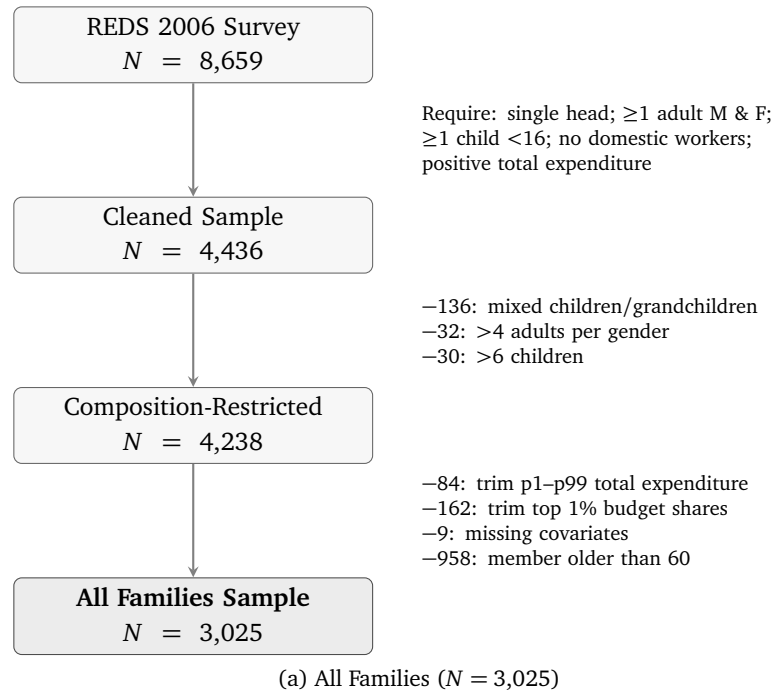
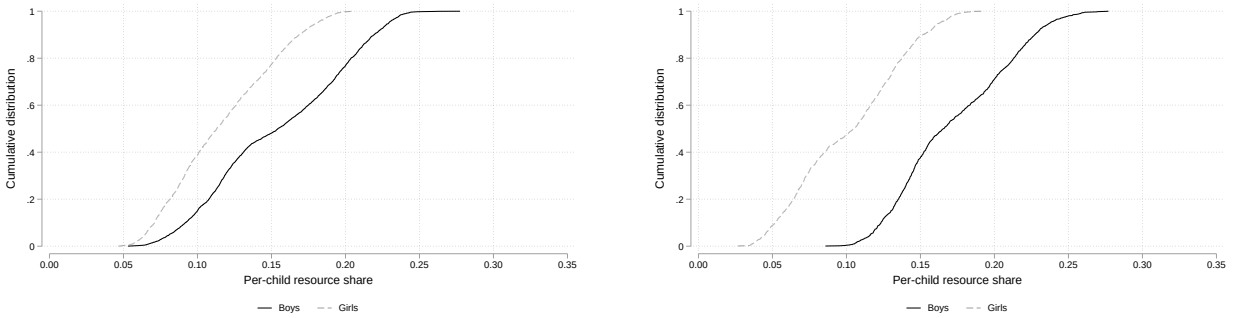


Figure A1: Sample Construction

Notes: Panel (a) traces the construction of the all-families analysis sample from the full REDS 2006 survey. Panel (b) traces the construction of the nuclear-families subsample, starting from the cleaned sample in Panel (a). Trimming uses expenditure and budget shares before excluding durable-repair expenses to preserve the fixed estimation samples. See Section 2.4 for details.

Figure A2: Distribution of Children's Resource Shares by Gender

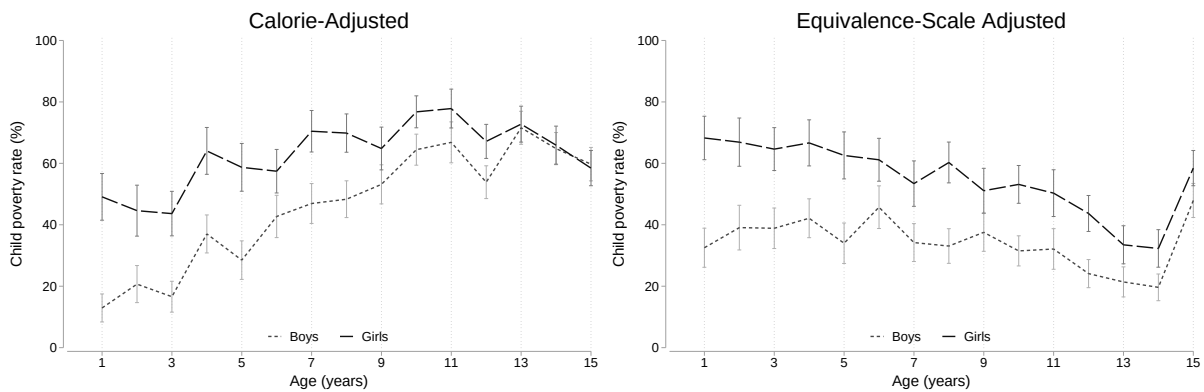


A) All Families

B) Nuclear Families

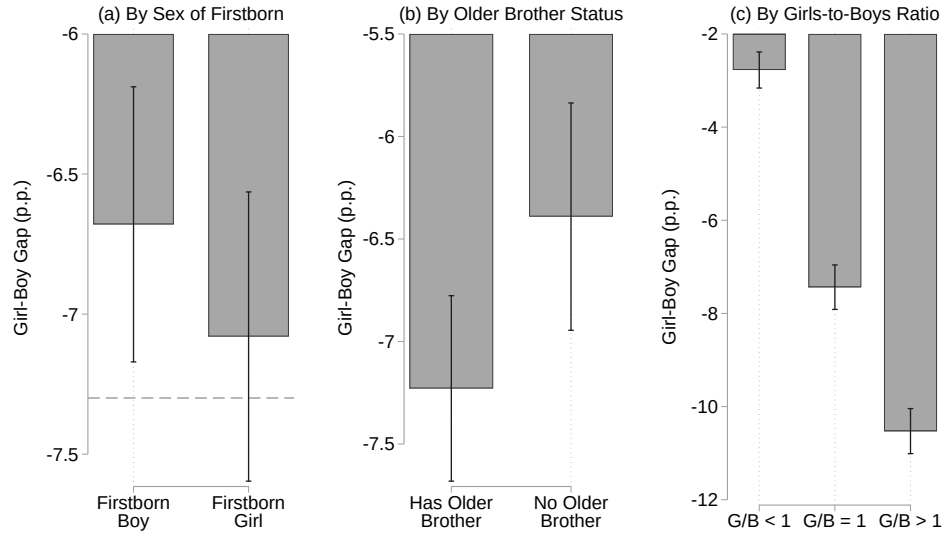
Notes: The panels plot empirical cumulative distribution functions of per-child resource shares for boys and girls in all families ($N=3,025$) and nuclear families ($N=1,410$). In both samples, the girls' CDF lies above the boys' CDF over most of the support, indicating systematically lower resource shares for girls. Kolmogorov-Smirnov tests reject equality of the two distributions in both samples (all families: $D = 0.335$; nuclear families: $D = 0.590$; $p < 0.01$ in both cases).

Figure A3: Child Poverty by Age



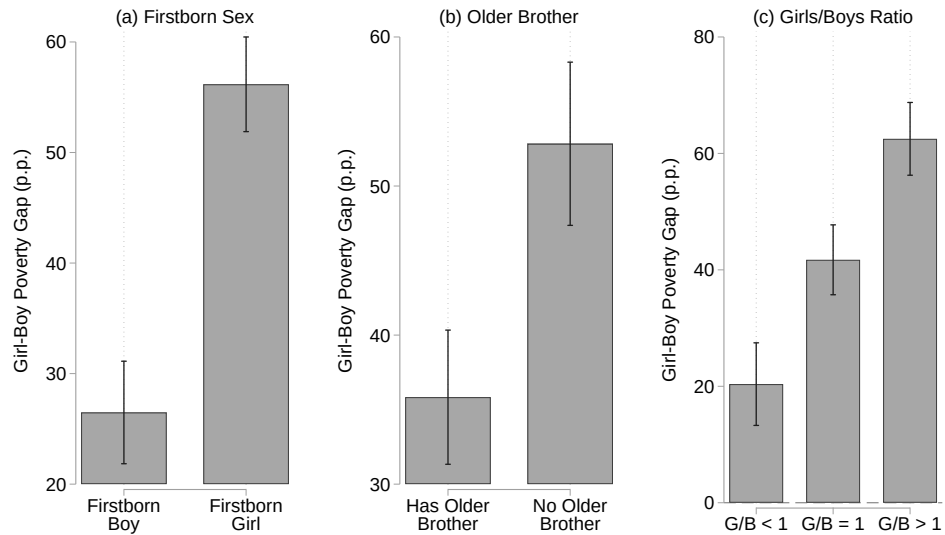
Notes: All families sample. The plotted age range is 1–15. In each panel, boys and girls are shown separately, with 95% confidence intervals at each age. The calorie-adjusted panels scale the adult World Bank benchmark (\$1.90/day) by Indian age-sex-specific calorific requirements from [Indian Council of Medical Research and National Institute of Nutrition \(2020\)](#); the equivalence-scale-adjusted panels assign children under age 15 a poverty threshold equal to 60 percent of the adult World Bank benchmark.

Figure A4: Girl-Boy Per-Child Resource-Share Gap by Family Composition



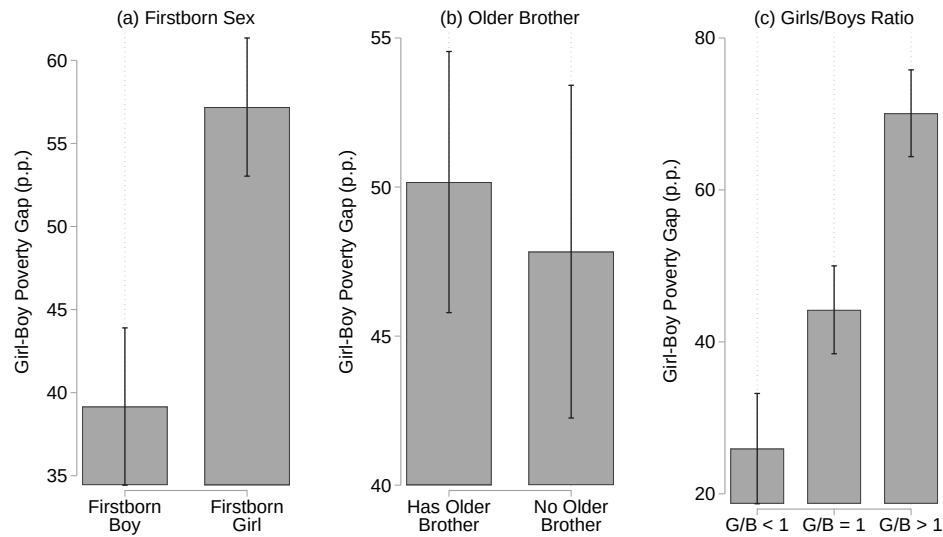
Notes: Sample: nuclear families with both boys and girls. Each bar shows the girl-boy gap in per-child resource shares, measured in percentage points, with 95% confidence intervals. Negative values indicate girls receive less per child. Between-group tests. The gap differs significantly by sex of firstborn and girls-to-boys ratio, but not by older brother status.

Figure A5: Girl-Boy Calorie-Adjusted Poverty Gap by Family Composition



Notes: Sample: nuclear families. Each bar shows the girl-boy gap in individual poverty rates, measured in percentage points, with 95% confidence intervals under a calorie-adjusted poverty measure that scales the adult World Bank benchmark (\$1.90/day) by Indian age-sex-specific calorific requirements from [Indian Council of Medical Research and National Institute of Nutrition \(2020\)](#). Positive values indicate higher poverty among girls. The gap is largest in firstborn-girl households and when girls outnumber boys.

Figure A6: Girl-Boy Equivalence-scale Adjusted Poverty Gap by Family Composition



Notes: Sample: nuclear families. Each bar shows the girl–boy gap in equivalence-scale-adjusted individual poverty rates, measured in percentage points, with 95% confidence intervals. The measure assigns children under age 15 a poverty threshold equal to 60 percent of the adult World Bank benchmark (\$1.90/day). Positive values indicate higher poverty among girls. The gap is largest in firstborn-girl households and when girls outnumber boys.

C Additional Tables

Table A3: Poverty Lines by Age and Sex in 2005–06 Rupees

Age (years)	Energy (kcal/day)		Calorie-adjusted (Rs./month)		Equivalence scale (Rs./month)
	Male	Female	Male	Female	
Under 1	595	595	112.24	112.24	271.64
1–3	1070	1070	201.84	201.84	271.64
4–6	1360	1360	256.55	256.55	271.64
7–9	1700	1700	320.68	320.68	271.64
10–12	2220	2060	418.77	388.59	271.64
13–14	2860	2400	539.50	452.73	271.64
15	2860	2400	539.50	452.73	452.73
16–18	3320	2500	626.27	471.59	452.73
19–46	2710	2130	511.20	401.80	452.73
47–60	2710	2130	511.20	401.80	362.18

Notes: Poverty lines are per person in 2005–06 rupees. The adult benchmark is the World Bank \$1.90/day line at 2011 PPP, converted using the rural India PPP factor of 13.173 and the CPI-AL ratio 358/602. Monthly lines equal annual lines divided by 12. Calorie-adjusted lines scale the benchmark by age- and sex-specific calorific requirements from [Indian Council of Medical Research and National Institute of Nutrition \(2020\)](#) relative to 2,400 kcal/day; age zero uses 595 kcal/day. Equivalence-scale-adjusted lines apply factors of 0.6 below age 15, 1 at ages 15–46, and 0.8 above age 46, equally for both sexes. Children use their own age; adults use the integer household mean age of adults of the same sex. Conventional household poverty uses the unscaled per-capita benchmark.

Table A4: Resource-Share Function Coefficients

	Nuclear Families				All Families			
	Men	Women	Boys	Girls	Men	Women	Boys	Girls
Constant	0.5131 ^{***} (0.1054)	0.2229 ^{***} (0.0668)	0.1407 ^{**} (0.0675)	0.1233 ^{**} (0.0492)	0.5362 ^{***} (0.1245)	0.1957 ^{**} (0.0991)	0.2140 ^{***} (0.0764)	0.0541 (0.0445)
North	-0.0915 ^{***} (0.0316)	0.0650 [*] (0.0343)	0.0154 (0.0231)	0.0111 (0.0134)	-0.0192 (0.0300)	0.0016 (0.0315)	0.0069 (0.0190)	0.0107 (0.0127)
Men's educ.	0.0003 (0.0024)	-0.0013 (0.0019)	0.0026 (0.0022)	-0.0017 (0.0016)	0.0001 (0.0030)	-0.0004 (0.0029)	0.0008 (0.0028)	-0.0005 (0.0014)
Women's educ.	-0.0030 (0.0031)	0.0019 (0.0026)	0.0009 (0.0023)	0.0002 (0.0022)	-0.0001 (0.0046)	-0.0003 (0.0038)	-0.0015 (0.0032)	0.0019 (0.0020)
Men's avg. age	0.0007 (0.0030)	0.0004 (0.0022)	-0.0027 (0.0023)	0.0015 (0.0018)	-0.0008 (0.0022)	-0.0003 (0.0018)	0.0002 (0.0014)	0.0009 (0.0009)
Women's avg. age	-0.0005 (0.0034)	0.0020 (0.0027)	0.0011 (0.0026)	-0.0026 (0.0019)	0.0004 (0.0025)	0.0024 (0.0022)	-0.0019 (0.0014)	-0.0009 (0.0010)
Boys' avg. age	-0.0014 (0.0031)	-0.0014 (0.0021)	0.0021 (0.0024)	0.0008 (0.0015)	-0.0014 (0.0030)	-0.0013 (0.0024)	0.0046 ^{**} (0.0021)	-0.0019 (0.0016)
Girls' avg. age	-0.0032 (0.0029)	-0.0019 (0.0019)	-0.0001 (0.0020)	0.0053 ^{**} (0.0021)	-0.0048 [*] (0.0028)	-0.0006 (0.0023)	-0.0007 (0.0019)	0.0061 ^{***} (0.0018)
No. adult men	—	—	—	—	0.0037 (0.0242)	-0.0012 (0.0208)	-0.0063 (0.0151)	0.0038 (0.0101)
No. adult women	—	—	—	—	0.0080 (0.0237)	0.0146 (0.0233)	-0.0142 (0.0137)	-0.0084 (0.0095)
No. boys	-0.0400 ^{***} (0.0147)	-0.0182 [*] (0.0107)	0.0783 ^{***} (0.0133)	-0.0201 ^{**} (0.0086)	-0.0205 (0.0184)	-0.0076 (0.0128)	0.0332 ^{**} (0.0165)	-0.0052 (0.0074)
No. girls	0.0156 (0.0154)	0.0015 (0.0113)	-0.0238 ^{**} (0.0099)	0.0067 (0.0115)	-0.0201 (0.0166)	0.0033 (0.0120)	-0.0184 ^{**} (0.0093)	0.0353 ^{***} (0.0117)
SC/ST/OBC	-0.0321 (0.0270)	-0.0031 (0.0187)	0.0236 (0.0188)	0.0116 (0.0147)	-0.0053 (0.0349)	-0.0056 (0.0312)	0.0048 (0.0207)	0.0061 (0.0149)
Hindu	-0.0230 (0.0393)	0.0153 (0.0321)	0.0143 (0.0217)	-0.0066 (0.0172)	-0.0278 (0.0475)	0.0273 (0.0398)	-0.0002 (0.0226)	0.0007 (0.0161)
Observations	1,410				3,025			

Notes: Nonlinear SUR estimates under the Similar Across People (SAP) restriction. Entries are coefficients δ_j in the group-allocation function $s_{jh} = N_{jh}\eta_{jh} = \mathbf{x}'_h\delta_j$. The corresponding per-person share is $\eta_{jh} = s_{jh}/N_{jh}$ for a present group. The men's column gives the residual coefficients when both boys and girls are present; an absent child group is omitted from the residual allocation. Village-clustered standard errors are in parentheses. Adult counts are fixed at one in nuclear families, so their coefficients are not identified and are shown as dashes. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A5: Kolmogorov-Smirnov Tests: Boys' vs Girls' Resource Share Distributions

Sample	D-statistic	p-value	N
All Families	0.335	<0.001	3,025
Nuclear Families	0.590	<0.001	1,410

Notes: Two-sample Kolmogorov-Smirnov tests comparing boys' and girls' per-child resource-share distributions, using each sample's own DLP estimates. Each household containing the relevant child type contributes one observation. All families: 2,465 households with boys and 2,037 with girls. Nuclear families: 1,201 with boys and 1,041 with girls. Reported p -values are conventional two-sample K-S diagnostics; they do not account for paired observations, village dependence, or estimation of resource shares.

Table A6: Robustness: Resource Shares in Households with Both Boys and Girls

	All Families			Nuclear Families		
	Mean	Median	SD	Mean	Median	SD
Men (η_m)	0.204	0.213	0.069	0.305	0.302	0.072
Women (η_w)	0.305	0.284	0.138	0.307	0.305	0.053
Boys (η_b)	0.154	0.148	0.044	0.173	0.163	0.046
Girls (η_g)	0.125	0.120	0.032	0.121	0.110	0.046
Girl-Boy Gap ($100 \times (\eta_g - \eta_b)$)		-2.9			-5.1	
p -value for gap = 0		0.168			0.026	
Observations		1,463			822	

Notes: Nonlinear SUR estimates re-estimated separately for all and nuclear families with at least one boy and one girl younger than 16. Rows report means, medians, and standard deviations of per-person shares, obtained by dividing each household's group allocation by its number of members of that type. Households containing that type receive equal weight. The girl-boy gap is the difference between these conditional mean per-child shares, in percentage points. Its two-sided p -value uses the delta method and the village-clustered covariance matrix of the fitted coefficients, treating household covariates and counts as fixed.

Table A7: Robustness: Resource Shares in Households with Exactly One Boy and One Girl

	All Families			Nuclear Families		
	Mean	Median	SD	Mean	Median	SD
Men (η_m)	0.333	0.381	0.105	0.373	0.381	0.047
Women (η_w)	0.225	0.265	0.069	0.293	0.278	0.035
Boys (η_b)	0.185	0.191	0.024	0.205	0.204	0.021
Girls (η_g)	0.131	0.131	0.025	0.130	0.130	0.023
Girl-Boy Gap ($100 \times (\eta_g - \eta_b)$)		-5.4			-7.6	
p -value for gap = 0		0.014			0.004	
Observations		607			317	

Notes: Households with exactly one boy and one girl younger than 16. Predictions use the estimates in Table 2. Rows report means, medians, and standard deviations of per-person shares, obtained by dividing each household's group allocation by its number of members of that type. Households containing that type receive equal weight. The girl-boy gap is the difference between these conditional mean per-child shares, in percentage points.

Table A8: Child Resource Shares in Boys-Only and Girls-Only Households

Household type	All Families		Nuclear Families	
	Households	Per child (%)	Households	Per child (%)
One boy only	559	20.64	144	22.62
One girl only	371	15.59	113	14.19
Two boys only	331	12.10	171	15.33
Two girls only	137	9.13	61	7.40

Notes: Households have exactly one or two children younger than 16, all boys or all girls. Entries are mean per-child resource shares as percentages of total household resources. Shares use the full-covariate estimates in Table 2 for each sample.

Table A9: Per-Person Resource Shares at Median Covariate Values

	All Families			Nuclear Families		
	Share (%)	SE	95% CI	Share (%)	SE	95% CI
Men	40.17	3.26	[33.78, 46.56]	41.38	3.71	[34.12, 48.64]
Women	27.89	3.05	[21.91, 33.87]	26.78	3.50	[19.92, 33.65]
Boys	19.78	2.09	[15.69, 23.88]	20.03	2.57	[14.99, 25.07]
Girls	12.15	1.73	[8.76, 15.54]	11.81	2.48	[6.95, 16.67]
Girl-boy gap	-7.63	2.58	[-12.69, -2.58]	-8.22	3.51	[-15.10, -1.34]
<i>p</i> -value for gap = 0		0.003			0.019	

Notes: Nonlinear SUR predictions evaluated at each covariate's sample median, including categorical indicators and member counts. Per-person shares are percentages of household resources; the girl-boy gap is in percentage points. Both reference profiles have one man, one woman, one boy, and one girl, so group and per-person shares coincide.

Table A10: Resource Shares: Demographics and Region

	All Families			Nuclear Families		
	Mean	Median	SD	Mean	Median	SD
Men (η_m)	0.341	0.354	0.126	0.388	0.390	0.081
Women (η_w)	0.227	0.266	0.072	0.313	0.312	0.038
Boys (η_b)	0.155	0.154	0.045	0.166	0.163	0.030
Girls (η_g)	0.115	0.108	0.035	0.095	0.095	0.034
Girl-Boy Gap ($100 \times (\eta_g - \eta_b)$)		-4.1			-7.1	
<i>p</i> -value for gap = 0		0.024			0.002	
Observations		3,025			1,410	

Notes: Nonlinear SUR estimates using the same samples as Table 2. Covariates are member counts, boys' and girls' average ages, and the North indicator. Adult counts are fixed at one in nuclear families and absorbed by the intercept. Rows report means, medians, and standard deviations of per-person shares, obtained by dividing each household's group allocation by its number of members of that type. Households containing that type receive equal weight. The girl-boy gap is the difference between these conditional mean per-child shares, in percentage points. Its two-sided *p*-value uses the delta method and the village-clustered covariance matrix of the fitted coefficients, treating household covariates and counts as fixed.

Table A11: Resource Shares in Households without Teenage Children

	All Families			Nuclear Families		
	Mean	Median	SD	Mean	Median	SD
Men (η_m)	0.364	0.390	0.133	0.401	0.399	0.087
Women (η_w)	0.221	0.260	0.071	0.303	0.302	0.037
Boys (η_b)	0.149	0.152	0.044	0.179	0.176	0.040
Girls (η_g)	0.108	0.107	0.029	0.098	0.104	0.033
Girl-Boy Gap ($100 \times (\eta_g - \eta_b)$)		-4.2			-8.0	
p -value for gap = 0		0.019			< 0.001	
Observations		1,652			909	

Notes: Households have no child aged 13–15. Shares use the full-covariate estimates in Table 2 for each sample. Rows report means, medians, and standard deviations of per-person resource shares, conditional on the member type being present. The girl–boy gap is the difference between mean per-child shares, in percentage points.

Table A12: Resource Shares Excluding Private-School Households

	All Families			Nuclear Families		
	Mean	Median	SD	Mean	Median	SD
Men (η_m)	0.344	0.358	0.129	0.384	0.368	0.089
Women (η_w)	0.230	0.267	0.075	0.299	0.297	0.032
Boys (η_b)	0.151	0.151	0.045	0.180	0.173	0.045
Girls (η_g)	0.114	0.110	0.036	0.105	0.105	0.039
Girl-Boy Gap ($100 \times (\eta_g - \eta_b)$)		-3.6			-7.5	
p -value for gap = 0		0.057			<0.001	
Observations		2,706			1,256	

Notes: Nonlinear SUR estimates excluding households with any enrolled child younger than 16 attending a nonprofit or for-profit private school. The main-specification covariates are retained. Rows report means, medians, and standard deviations of per-person shares, obtained by dividing each household's group allocation by its number of members of that type. Households containing that type receive equal weight. The girl–boy gap is the difference between these conditional mean per-child shares, in percentage points. Its two-sided p -value uses the delta method and the village-clustered covariance matrix of the fitted coefficients, treating household covariates and counts as fixed.

Table A13: Robustness: Resource Shares under Alternative Household-Sample Restrictions

Sample	Children under 16	Adult structure	Observations	η_m	η_w	η_b	η_g	Gap	p-value
All families	Children or grandchildren of head; mixed child-grandchild households excluded	Other currently allowed adults may remain	3,025	0.344	0.225	0.155	0.117	-3.8	0.036
Nuclear families	Head's children only	Head and spouse only; no other adults or elderly	1,410	0.390	0.302	0.172	0.102	-7.0	<0.001
Head and spouse present; head's children only	Head's children only; no under-16 grandchildren	Spouse of head present; other currently allowed adults may remain	2,353	0.369	0.247	0.156	0.127	-2.9	0.202
Head's children only; extended adults allowed	Head's children only; no under-16 grandchildren	No spouse requirement; other currently allowed adults may remain	2,413	0.368	0.247	0.155	0.128	-2.7	0.190
Broad kinship sample	Any resident kin under 16; explicit non-relatives excluded	Production relationship restrictions relaxed; adult siblings and other relatives may remain	3,441	0.341	0.207	0.151	0.123	-2.8	0.086

Notes: Each resource share is the allocation to one member of the indicated type: the estimated group allocation is divided by the number of members of that type within each household. Summary statistics give equal weight to households containing that type. The first two rows report the main all-family and nuclear-family estimates; the remaining rows re-estimate the DLP model on the indicated samples. All samples exclude households with members older than 60. Resource shares are mean per-person shares, conditional on the group being present. The reported gap is girls minus boys in percentage points. The p-value tests H_0 : average $\eta_g - \eta_b = 0$ with village-clustered standard errors. Other allowed adults comprise the household head, spouse, children, and daughters-in-law aged 16–60. The broad kinship sample also includes other relatives, but excludes non-relatives, live-in servants or laborers, second spouses, and households with multiple heads.

Table A14: Resource Share Estimates Including Elderly Members

	All Families		
	Mean	Median	SD
Men (η_m)	0.304	0.323	0.113
Women (η_w)	0.288	0.336	0.097
Boys (η_b)	0.115	0.107	0.038
Girls (η_g)	0.084	0.079	0.026
Elderly (η_e)	0.153	0.158	0.031
Girl-Boy Gap ($100 \times (\eta_g - \eta_b)$)		-3.1	
p -value for gap = 0		0.189	
Observations		3,983	

Notes: Nonlinear SUR estimates for 3,983 families, including 958 with members older than 60. Adults aged 16–60 enter the male and female groups; members older than 60 form a fifth group matching the pooled elderly clothing category. Separate counts of elderly men and women and their average age enter alongside the baseline covariates. Rows report means, medians, and standard deviations of per-person shares, obtained by dividing each household's group allocation by its number of members of that type. Households containing that type receive equal weight. The girl–boy gap is the difference between these conditional mean per-child shares, in percentage points. Its two-sided p -value uses the delta method and the village-clustered covariance matrix of the fitted coefficients, treating household covariates and counts as fixed.

Table A15: GE(0) Decomposition of Inequality in Child Consumption

Sample	Total GE(0)	Between-HH	Within-HH	Within Share
All children	0.1493	0.1337	0.0155	10.4%
Nuclear families	0.1646	0.1224	0.0423	25.7%
Households with boys and girls	0.1336	0.1097	0.0239	17.9%

Notes: GE(0) is the mean log deviation of estimated child consumption, with equal weight for each child. Total inequality is decomposed into between-household and within-household components. Nuclear-family estimates use the nuclear-family DLP model; other rows use the all-family model. The within-household component captures differences between boys and girls, since consumption is assumed equal among children of the same sex within each household. It therefore provides a lower bound on within-household inequality among children.

Table A16: Individual Poverty Rates

	Calorie-Adjusted	Equivalence-Scale	Planning Commission
Men	21.7%	15.0%	18.2%
Women	31.6%	40.1%	46.2%
Boys	48.4%	33.5%	38.5%
Girls	63.6%	53.8%	59.5%
Girl-Boy Gap (p.p.)	15.2	20.3	21.0
Observations	3,025	3,025	3,025

Notes: Individual consumption equals the estimated gender-group resource share divided by its group size, multiplied by total household expenditure. The first column reports a calorie-adjusted poverty measure that scales the adult World Bank benchmark (\$1.90/day) by age- and sex-specific calorific requirements from [Indian Council of Medical Research and National Institute of Nutrition \(2020\)](#). The second column reports an equivalence-scale-adjusted poverty measure that assigns children under age 15 a threshold equal to 60 percent of the adult benchmark. The third column uses the Planning Commission's rural poverty line of Rs. 13 per day in 1999 prices, adjusted to fixed 2005–06 prices. The sample includes both nuclear and extended households. The reported gap is girls minus boys in percentage points, so positive values indicate higher poverty among girls.

Table A17: Calorie-Adjusted and Equivalence-Scale-Adjusted Hidden Poverty by Household Poverty Status

	Calorie-Adjusted			Equivalence-Scale Adjusted		
	Girls	Boys	Girl-Boy Gap (p.p.)	Girls	Boys	Girl-Boy Gap (p.p.)
<i>All families</i> (N=3,025; HH non-poor=2,005; HH poor=1,020)						
HH Non-Poor	42.3%	28.8%	13.6	28.9%	8.5%	20.3
HH Poor	93.8%	79.9%	14.0	89.3%	73.5%	15.7

Notes: Values are child poverty rates within each household-poverty stratum. Household poverty status is computed using household per-capita expenditure and the adult World Bank benchmark (\$1.90/day). The left block reports a calorie-adjusted poverty measure that scales the adult benchmark by Indian age- and sex-specific calorific requirements from [Indian Council of Medical Research and National Institute of Nutrition \(2020\)](#); the right block reports an equivalence-scale-adjusted poverty measure that assigns children under age 15 a threshold equal to 60 percent of the adult benchmark. The reported gap is girls minus boys in percentage points, so positive values indicate higher poverty among girls.

Table A18: Evidence of Son-Biased Fertility Stopping Behavior

	Firstborn Boy	Firstborn Girl	Difference
Number of children	2.35	2.55	0.19*** (0.06)
Fraction of younger siblings who are boys	0.516	0.632	0.115*** (0.024)
Observations	770	638	

Notes: Sample: nuclear families (N=1,408). Two households with tied firstborn ages (mixed-gender oldest children) are excluded. Difference column reports the firstborn-girl mean minus the firstborn-boy mean, with standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A19: Poverty Decomposition: Village-Jackknife Diagnostics

	Estimate	Jackknife SE	95% interval
<i>Panel A. Calorie-Adjusted</i>			
Total gap	39.32	1.78	[35.83, 42.80]
Within component	35.53	1.87	[31.86, 39.19]
Composition component	3.79	0.90	[2.03, 5.55]
<i>Panel B. Equivalence-Scale Adjusted</i>			
Total gap	48.16	2.15	[43.94, 52.38]
Within component	44.66	2.37	[40.02, 49.30]
Composition component	3.49	1.05	[1.44, 5.55]

Notes: Nuclear-family sample. Entries are percentage points. The table decomposes the girl-boy poverty gap into within-household-type and household-composition components, with household type defined by the sex of the first-born child. The within component uses boys' distribution across household types; the composition component captures the contribution of girls' and boys' different distributions. Each of the 234 villages is omitted once, and DLP poverty rates, and decomposition weights are re-estimated on the remaining households using the selected fitted coefficients as starting values. Standard errors use the village jackknife; intervals equal the original estimate plus or minus 1.96 standard errors.

Table A20: Individual Poverty and Child Human Capital Outcomes: Excluding Private-School Households

	Girls				Boys			
	Health	Human Capital			Health	Human Capital		
	HAZ	Education Spending	Advanced Reading	Advanced Writing	HAZ	Education Spending	Advanced Reading	Advanced Writing
<i>Panel A. Calorie-Adjusted Poverty Measure</i>								
Individually Poor	−0.071 (0.056)	−0.550*** (0.094)	−0.148*** (0.036)	−0.125*** (0.035)	−0.049 (0.058)	−0.339*** (0.066)	−0.062** (0.031)	−0.041 (0.032)
HH Per-Capita Poor	−0.046 (0.054)	−0.347*** (0.083)	−0.016 (0.037)	−0.017 (0.034)	−0.083 (0.054)	−0.423*** (0.070)	−0.037 (0.036)	−0.058 (0.036)
Observations	2,572	1,469	1,488	1,488	3,228	2,023	2,037	2,037
<i>Panel B. Equivalence-Scale-Adjusted Poverty Measure</i>								
Individually Poor	−0.109* (0.062)	−0.451*** (0.100)	−0.035 (0.042)	−0.012 (0.039)	−0.092 (0.070)	−0.164** (0.064)	−0.004 (0.035)	−0.023 (0.036)
HH Per-Capita Poor	−0.019 (0.058)	−0.324*** (0.094)	−0.061 (0.042)	−0.065* (0.037)	−0.049 (0.063)	−0.498*** (0.076)	−0.066* (0.036)	−0.065* (0.038)
Observations	2,572	1,469	1,488	1,488	3,228	2,023	2,037	2,037

Notes: Sample excludes households with an enrolled child younger than 16 identified as attending a nonprofit or for-profit private school, leaving 2,706 all-family households. Households with unresolved school or enrollment information remain. Individual poverty uses DLP resource shares re-estimated on this sample and the poverty thresholds in Table 4. HAZ is internally standardized by age and sex, with the same trimming cutoffs as Table 4. Education Spending is $\ln(1 + \text{annual Rs.})$; Advanced Reading and Advanced Writing indicate ability in both English and the mother tongue. All specifications control for child age and numbers of boys, girls, men, and women in the household. Village-clustered standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A21: Robustness: Missing-Girls Counterfactuals

Specification	Fixed effects used	β	p-value	Share explained
<i>Panel A. Calorie-Adjusted Poverty Measure</i>				
Pooled, unweighted	None	6.45	0.003	47.4%
State FE, unweighted	State	8.12	0.003	55.4%
State + age FE, unweighted	State + age group	2.36	0.149	21.9%
Pooled, population-weighted	None	9.67	< 0.001	62.4%
State FE, population-weighted	State	11.66	< 0.001	71.1%
State + age FE, population-weighted	State + age group	3.95	0.055	33.0%
<i>Panel B. Equivalence-Scale-Adjusted Poverty Measure</i>				
Pooled, unweighted	None	3.79	0.089	32.9%
State FE, unweighted	State	5.96	0.066	44.0%
State + age FE, unweighted	State + age group	-0.14	0.933	-1.8%
Pooled, population-weighted	None	6.31	0.016	45.5%
State FE, population-weighted	State	8.66	0.015	55.7%
State + age FE, population-weighted	State + age group	2.01	0.271	20.2%

Notes: All specifications use rural SRS mortality rates and rural female populations by state and age group. Panel A uses calorie-adjusted girl-boy poverty gaps; Panel B uses equivalence-scale-adjusted gaps. Beta is the coefficient from a regression of excess female deaths per 1,000 girls on the poverty gap. Population-weighted specifications weight cells by their rural female population. Counterfactual missing-girl counts subtract $\hat{\beta}\Delta P_{sa}$ from each cell's excess female death rate, set negative adjusted rates to zero, and multiply by the corresponding female population. The explained share is the resulting reduction as a percentage of estimated missing girls.